INCH-POUND

MIL-M-38510/172C
04 June 2004

SUPERSEDING
MIL-M-38510/172B
30 April 1984

MILITARY SPECIFICATION
MICROCIRCUITS, DIGITAL, CMOS, AND-OR-INVERT, EXCLUSIVE-OR, EXCLUSIVE-NOR GATES, MONOLITHIC SILICON, POSITIVE LOGIC

Reactivated after 04 June 2004 and may be used for new and existing designs and acquisitions.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product herein shall consist of this specification sheet and MIL-PRF 38535.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, CMOS, logic microcircuits. Two product assurance classes and a choice of case outlines, lead finishes, and radiation hardness assurance (RHA) are provided and are reflected in the complete Part or Identifying Number (PIN). For this product, the requirements of MIL-M-38510 have been superseded by MIL-PRF-38535 (see 6.3).

1.2 Part or identifying number (PIN). The PIN is in accordance with MIL-PRF-38535 and as specified herein.

1.2.1 Device types. The device types are as follows:

Device type	Circuit
01	Dual 2-wide, 2-input AND-OR-invert gate
02	Expandable 4-wide, 2-input AND-OR invert gate
03	Quad exclusive-OR gate
04	Quad exclusive-NOR gate

1.2.2 Device class. The device class is the product assurance level as defined in MIL-PRF-38535.

1.2.3 Case outlines. The case outlines are as designated in MIL-STD-1835 and as follows:

Outline letter	Descriptive designator	Terminals	Package style
A	GDFP5-F14 or CDFP6-F14	14	Flat pack
C	GDIP1-T14 or CDIP2-T14	14	Dual-in-line
D	GDFP1-F14 or CDFP2-F14	14	Flat pack
T	CDFP3-F14	14	Flat pack
X 1/ 2/	GDFP5-F14 or CDFP6-F14	14	Flat pack, except A dimension equals 0.100” (2.54 mm) max
Y 1/ 2/	GDFP1-F14 or CDFP2-F14	14	Flat pack, except A dimension equals 0.100” (2.54 mm) max

1/ As an exception to nickel plate or undercoating paragraph of MIL-PRF-38535, appendix A; for case outlines X and Y only, the leads of bottom brazed ceramic packages (i.e., configuration 2 of case outline A or D) may have electroless nickel undercoating which shall be 50 to 200 microinches (1.27 to 5.08 µm) thick provided the lead finish is hot solder dip (i.e., finish letter A) and provided that, after any lead forming, an additional hot solder dip coating is applied which shall extend from the outer tip of the lead to no more than 0.015 inch (0.38 mm) from the package edge.

2/ For bottom or side brazed packages, case outlines X and Y only, the S_1 dimension may go to .000 inch (.00 mm) minimum.

Comments, suggestions, or questions on this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAC, P.O Box 3990, Columbus, OH 43218-3990, or email CMOS@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at www.dodssp.daps.mil.

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1.3 Absolute maximum ratings.

Supply voltage range (V_{DD} - V_{SS})	-0.5 V dc to +18.0 V dc
Input current (each input)	±10 mA
Input voltage range	(V_{SS} - 0.5) ≤ V_I ≤ (V_{DD} + 0.5)
Storage temperature range (T_{STG})	-65° to +175°C
Maximum power dissipation (P_D)	200 mW
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction to case (Θ_{JC})	See MIL-STD-1835
Junction temperature (T_J)	175°C

1.4 Recommended operating conditions.

Supply voltage range (V_{DD} - V_{SS})	4.5 V dc to 15.0 V dc
Input low voltage range (V_{IL})	V_{OL} = 10% V_{DD}, V_{OH} = 90% V_{DD} 0.0 V to 1.5 V dc at V_{DD} = 5.0 V dc 0.0 V to 2.0 V dc at V_{DD} = 10.0 V dc 0.0 V to 4.0 V dc at V_{DD} = 15.0 V dc
Input high voltage range (V_{IH})	V_{OL} = 10% V_{DD}, V_{OH} = 90% V_{DD} 3.5 V to 5.0 V dc at V_{DD} = 5.0 V dc 8.0 V to 10.0 V dc at V_{DD} = 10.0 V dc 11 V to 15.0 V dc at V_{DD} = 15.0 V dc
Ambient operating temperature range (T_A)	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and Standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATION

MIL-PRF-38535 - Integrated Circuits (Microcircuits) Manufacturing, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-883 - Test Method Standard Microcircuits.
MIL-STD-1835 - Interface Standard Electronic Component Case Outlines.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or www.dodssp.daps.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. Microcircuits furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.3 and 6.4).

3.2 Item requirements. The individual item requirements shall be in accordance with MIL-PRF-38535 and as specified herein or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not affect the form, fit, or function as described herein.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-PRF-38535 and herein. Although eutectic die bonding is preferred, epoxy die bonding may be performed. However the resin used shall be Dupont 5504 Conductive Silver Paste, or equivalent, which is cured at 200°C ±10°C for a minimum of 2 hours. The use of equivalency epoxies or cure cycles shall be approved by the qualifying activity. Equivalency shall be demonstrated in data submitted to the qualifying activity for verification.

3.3.1 Logic diagrams and terminal connections. The logic diagrams and terminal connections shall be as specified on figure 1.

3.3.2 Truth tables and logic equations. The truth tables and logic equations shall be as specified on figure 2.

3.3.3 Switching waveforms and test circuit. The switching waveforms and test circuit shall be as specified on figure 3.

3.3.4 Schematic circuits. The schematic circuits shall be maintained by the manufacturer and made available to the qualifying activity or preparing activity upon request.

3.3.5 Case outlines. The case outlines shall be as specified in 1.2.3.

3.4 Lead material and finish. The lead material and finish shall be in accordance with MIL-PRF-38535 (see 6.6).

3.5 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics are as specified in table I, and apply over the full recommended ambient operating temperature range.

3.6 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.7 Marking. Marking shall be in accordance with MIL-PRF-38535.

3.7.1 Radiation hardness assurance identifier. The radiation hardness assurance identifier shall be in accordance with MIL-PRF-38535 and 4.5.4 herein.

3.8 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 36 (see MIL-PRF-38535, appendix A).

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions V_{SS} = 0 V -55°C ≤ T_A ≤ +125°C Unless otherwise specified		Device type	Limits		Unit
					Min	Max	
Positive clamping input to V_{DD}	$V_{IC\ (POS)}$	T_A = +25°C, V_{DD} = GND V_{SS} = Open, Output = Open I_{IN} = 1 mA		All		1.5	V dc
Negative clamping input to V_{SS}	$V_{IC\ (NEG)}$	T_A = +25°C, V_{DD} = GND V_{SS} = GND, Output = Open I_{IN} = -1 mA		All		-6.0	V dc
Quiescent supply current	I_{SS} or I_{DD}	V_{DD} = 18 V dc V_{IN} = V_{SS} or V_{DD} All input combinations		All		-750	nA
High level output voltage	V_{OH1}	V_{DD} = 15 V dc Any one input = V_{DD} All other inputs = V_{SS} $\lvert I_O \rvert$ ≤ 1µA		01, 02, 03	14.95		V dc
		V_{DD} = 15 V dc V_{IN} = V_{SS} V_{IN} = V_{DD} $\lvert I_O \rvert$ ≤ 1µA		04	14.95		
Low level output voltage	V_{OL1}	V_{DD} = 15 V dc V_{IN} = V_{DD} $\lvert I_O \rvert$ ≤ 1µA		01, 02		0.05	V dc
		V_{DD} = 15 V dc V_{IN} = V_{DD} V_{IN} = V_{SS} $\lvert I_O \rvert$ ≤ 1µA		03		0.05	V dc
		V_{DD} = 15 V dc Any one input = V_{DD} All other inputs = V_{SS} $\lvert I_O \rvert$ ≤ 1µA		04		0.05	V dc
Input high voltage	V_{IH1}	V_{DD} = 5 V dc Any one input = 3.5 V dc All other inputs = V_{SS} $\lvert I_O \rvert$ ≤ 1µA	V_O = 4.5 Vdc	01	3.5		V dc
			V_O = 0.5 Vdc	02			
			V_O = 4.5 Vdc	03			
			V_O = 0.5 Vdc	04			
	V_{IH2}	V_{DD} = 10 V dc Any one input = 7.0 V dc All other inputs = V_{SS} $\lvert I_O \rvert$ ≤ 1µA	V_O = 9.0 Vdc	01	7.0		V dc
			V_O = 1.0 Vdc	02			
			V_O = 9.0 Vdc	03			
			V_O = 1.0 Vdc	04			
	V_{IH3}	V_{DD} = 15 V dc Any one input = 11.0 V dc All other inputs = V_{SS} $\lvert I_O \rvert$ ≤ 1µA	V_O = 13.5 Vdc	01	11.0		V dc
			V_O = 1.5 Vdc	02			
			V_O = 13.5 Vdc	03			
			V_O = 1.5 Vdc	04			

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TABLE I. Electrical performance characteristics – Continued.

Test	Symbol	Conditions V_{SS} = 0 V -55°C ≤ T_A ≤ +125°C Unless otherwise specified		Device type	Limits		Unit
					Min	Max	
Input low voltage	V_{IL1}	V_{DD} = 5 V dc Any one input = 1.5 V All other inputs = V_{SS} $\|I_O\| \leq 1\mu A$	V_O = 4.5 Vdc	01, 04		1.5	V dc
			V_O = 0.5 Vdc	02, 03			
	V_{IL2}	V_{DD} = 10 V dc Any one input = 3.0 V All other inputs = V_{SS} $\|I_O\| \leq 1\mu A$	V_O = 9.0 Vdc	01, 04		3.0	V dc
			V_O = 1.0 Vdc	02, 03			
	V_{IL3}	V_{DD} = 15 V dc Any one input = 4.0 V All other inputs = V_{SS} $\|I_O\| \leq 1\mu A$	V_O = 13.5 Vdc	01, 04		4.0	V dc
			V_O = 1.5 Vdc	02, 03			
Output high (source) current	I_{OH1}	V_{DD} = 5.0 V dc V_{IN} = V_{SS} V_{OH} = 4.6 V dc		01, 02, 04	-0.36		mA dc
		V_{DD} = 5.0 V dc Any one input = V_{DD} All other inputs = V_{SS} V_{OH} = 4.6 V dc		03			
	I_{OH2}	V_{DD} = 15 V dc V_{IN} = V_{SS} V_{OH} = 13.5 V dc		01, 02, 04	-2.4		mA dc
		V_{DD} = 15 V dc Any one input = V_{DD} All other inputs = V_{SS} V_{OH} = 13.5 V dc		03			
Output low (sink) current	I_{OL1}	V_{DD} = 5.0 V dc V_{IN} = V_{SS} V_{OL} = 0.4 V dc		01, 02, 03	0.36		mA dc
		V_{DD} = 5.0 V dc Any one input = V_{DD} All other inputs = V_{SS} V_{OL} = 0.4 V dc		04			
	I_{OL2}	V_{DD} = 15 V dc V_{IN} = V_{SS} V_{OL} = 1.5 V dc		01, 02, 03	2.4		mA dc
		V_{DD} = 15 V dc Any one input = V_{DD} All other inputs = V_{SS} V_{OL} = 1.5 V dc		04			

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TABLE I. Electrical performance characteristics – Continued.

Test	Symbol	Conditions V_{SS} = 0 V -55°C ≤ T_A ≤ +125°C Unless otherwise specified	Device type	Limits		Unit
				Min	Max	
Input leakage 1/ current, high	I_{IH}	V_{DD} = 18 V dc Measure inputs sequentially; connect all unused inputs to V_{SS}	01, 02, 04		45	nA
			03		100	nA
Input leakage 1/ current, low	I_{IL}	V_{DD} = 18 V dc Measure inputs sequentially; connect all unused inputs to V_{DD}	01, 02, 04		-45	nA
			03		-100	nA
Input capacitance	C_i	V_{DD} = 0 V dc, f = 1 MHz, T_A = 25°C	All		7.5	pF
Propagation delay times, high to low level	t_{PHL}	V_{DD} = 5 V dc, C_L = 50 pF R_L = 200 kΩ (See figure 3)	01, 02	15	630	ns
			03, 04	13	350	
		V_{DD} = 10 V dc, C_L = 50 pF R_L = 200 kΩ, T_A = 25°C (See figure 3)	01, 02	6	180	ns
			03, 04	6	120	
Propagation delay times, low to high level	t_{PLH}	V_{DD} = 5 V dc, C_L = 50 pF R_L = 200 kΩ (See figure 3)	01	25	870	ns
			02	25	980	
			03, 04	13	350	
		V_{DD} = 10 V dc, C_L = 50 pF R_L = 200 kΩ, T_A = 25°C (See figure 3)	01	10	250	ns
			02	14	280	
			03, 04	6	120	
Transition times, high to low level	t_{THL}	V_{DD} = 5 V dc, C_L = 50 pF R_L = 200 kΩ (See figure 3)	All	10	280	ns
		V_{DD} = 10 V dc, C_L = 50 pF R_L = 200 kΩ T_A = +25°C (See figure 3)	All	5	100	ns
Transition times, low to high level	t_{TLH}	V_{DD} = 5 V dc, C_L = 50 pF R_L = 200 kΩ (See figure 3)	All	10	280	ns
		V_{DD} = 10 V dc, C_L = 50 pF R_L = 200 kΩ T_A = +25°C (See figure 3)	All	5	100	ns

1/ Input current of one input node.

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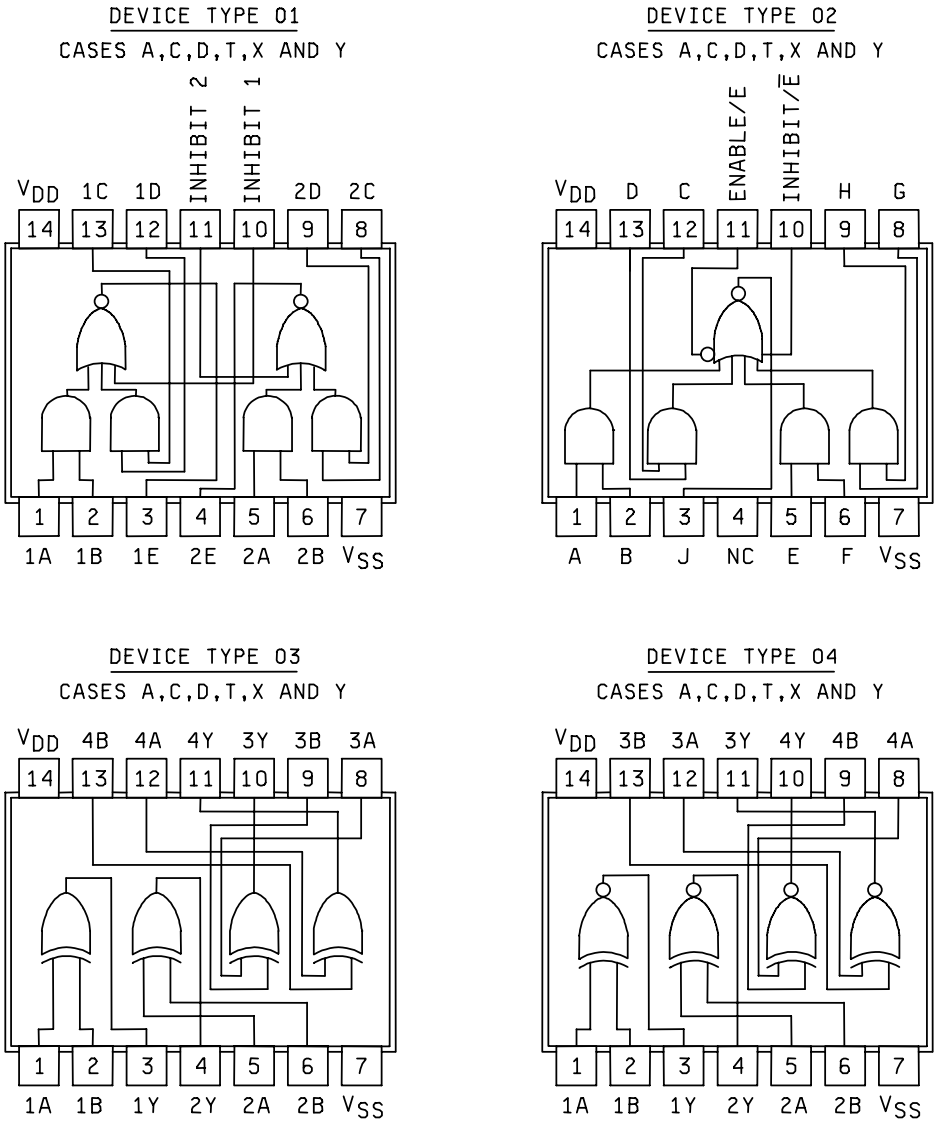


FIGURE 1. Logic diagrams and terminal connections.

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Device type 01

Truth table					
Inputs					Output
A	B	C	D	I	E
L	L	L	L	L	H
L	L	L	L	H	L
H	H	L	L	L	L
L	L	H	H	L	L
H	H	H	H	L	L
H	L	H	L	L	H
L	H	L	H	L	H

$E = \overline{\text{Inhibit} + AB + CD}$

Device type 02

Truth table										
Inputs										Output
A	B	C	D	E	F	G	H	I	E	J
L	L	L	L	L	L	L	L	L	H	H
L	L	L	L	L	L	L	L	H	H	L
L	L	L	L	L	L	L	L	L	L	L
H	H	H	H	H	H	H	H	L	H	L
H	H	L	L	L	L	L	L	L	H	L
L	L	H	H	L	L	L	L	L	H	L
L	L	H	H	L	L	L	L	L	H	L
L	L	L	L	H	H	L	L	L	H	L
L	L	L	L	L	L	H	H	L	H	L
H	L	H	L	H	L	H	L	L	H	H

$J = \overline{\text{Inhibit} + \text{Enable} + AB + CD + EF + GH}$

Device type 03

Truth table		
Inputs		Output
A	B	Y
L	L	L
L	H	H
H	L	H
H	H	L

$Y = A \oplus B$

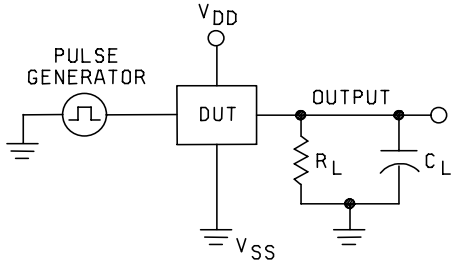
Device type 04

Truth table		
Inputs		Output
A	B	Y
L	L	H
L	H	L
H	L	L
H	H	H

$Y = \overline{A \oplus B}$

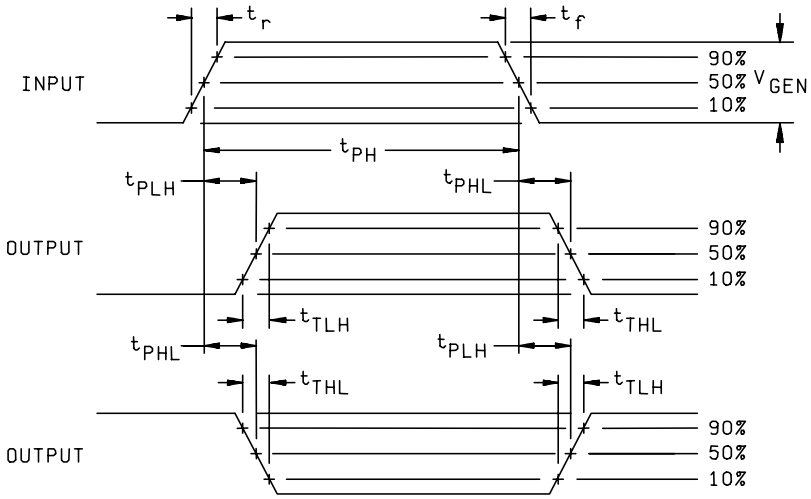
FIGURE 2. Truth tables and logic equations.

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R_L = 200 kΩ ±5%
C_L = 50 pF ±5%
INCLUDES WIRING AND PROBE CAPACITANCE

TEST CIRCUIT



DYNAMIC TEST WAVEFORMS

INPUT PULSE
V_{GEN} = V_{DD} ±1.0%
t_{PH} = 1.0 ±0.1 µs
t_r and t_f ≤ 20 ns
PRR = 200 kHz

NOTE: All unused inputs must be tied to V_{SS}.

FIGURE 3. Switching waveforms and test circuit.

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4. VERIFICATION

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-PRF-38535 or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not affect the form, fit, or function as described herein.

4.2 Screening. Screening shall be in accordance with, MIL-PRF-38535 and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. The burn-in test duration, test condition, and test temperature, or approved alternatives shall be as specified in the device manufacturer's QM plan in accordance with MIL-PRF-38535. The burn-in test circuit shall be maintained under document control by the device manufacturer's Technology Review Board (TRB) in accordance with MIL-PRF-38535 and shall be made available to the acquiring or preparing activity upon request. The test circuit shall specify the inputs, outputs, biases, and power dissipation, as applicable, in accordance with the intent specified in test method 1015 of MIL-STD-883.

b. Delete the sequence specified as initial (pre-burn-in) electrical parameters through interim (post-burn-in) electrical parameters of method 5004 and substitute lines 1 through 7 of table II herein.

c. Burn-in (method 1015 of MIL-STD-883).

(1) Unless otherwise specified in the manufacturers QM plan for static tests (test condition A), ambient temperature (T_A) shall be +125°C minimum. Test duration for each static test shall be 24 hours minimum for class S devices and in accordance with table I of method 1015 for class B devices.

i. For static burn-in I, all inputs shall be connected to 0.0 V.

ii. For static burn-in II, all inputs shall be connected to V_{DD}. V_{DD} = 15 V minimum and 18 V maximum.

iii. Except for V_{DD} and V_{SS}, terminals shall be connected through resistors which value is 2 kΩ to 47 kΩ. The actual measured value of the resistor selected shall not exceed ±20% of its branded value due to use, heat or age.

iv. Output may be open or connected to $V_{DD}/2$.

(2) Unless otherwise specified in the manufacturers QM plan for dynamic test (test condition D), ambient temperature shall be +125°C minimum. Test duration shall be in accordance with table I of method 1015.

i. Except for V_{DD} and V_{SS}, terminals shall be connected through resistors which value is 2 kΩ to 47 kΩ. The actual measured value of the resistor selected shall not exceed ±20% of its branded value due to use, heat or age.

ii. Input signal requirements: Square wave, 50% duty cycle; 25 kHz ≤ PRR ≤ 1 MHz; t_{TLH} and $t_{THL} \leq 1$ µs. Voltage level: Minimum = V_{SS} - 0.5 V, +10% V_{DD}; maximum = V_{DD} + 0.5 V, -10% V_{DD}.

iii. V_{DD} = 15.0 V minimum, 18.0 V maximum; $V_{DD}/2 = V_{DD}/2 \pm 1.0$ V; V_{SS} = 0.0 V.

d. Interim and final electrical test parameters shall be as specified in table II.

e. For class S devices, post dynamic burn-in, or class B devices, post static burn-in, electrical parameter measurements may, at the manufacturer's option, be performed separately or included in the final electrical parameter requirements.

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TABLE II. Electrical test requirements.

Line no.	MIL-PRF-38535 test requirements	Class S device 1/			Class B device 1/		
		Ref. par.	Table III Subgroups 2/	Table IV delta limits 3/	Ref. par.	Table III subgroups 2/	Table IV delta limits 3/
1	Interim electrical parameters		1			1	
2	Static burn-in I (method 1015)	4.2c 4.5.2					
3	Same as line 1		1	Δ			
4	Static burn-in II (method 1015)	4.2c 4.5.2			4.2c 4.5.2	4/	
5	Same as line 1	4.2e	1*	Δ	4.2e	1*	Δ
6	Dynamic burn-in (method 1015)	4.2c 4.5.2					
7	Same as line 1	4.2e	1*	Δ			
8	Final electrical parameters (method 5004)		1*, 2, 3, 9			1*, 2, 3, 9, 10, 11	
9	Group A test requirements (method 5005)	4.4.1	1, 2, 3, 4, 9, 10, 11, 12		4.4.1	1, 2, 3, 4, 9	
10	Group B test when using method 5005 QCI option	4.4.2	1, 2, 3, 9, 10, 11	Δ			
11	Group C end-point electrical parameters (method 5005)				4.4.3	1, 2,3	Δ
12	Group D end-point electrical parameters (method 5005)	4.4.4	1, 2, 3		4.4.4	1, 2, 3	

1/ Blank spaces indicate tests are not applicable.

2/ * indicates PDA applies to subgroup 1 (see 4.2.1).

3/ Δ indicates delta limits shall be required only on table III subgroup 1, where specified, and the delta values shall be computed with reference to the previous interim electrical parameters.

4/ The device manufacturer may at his option either perform delta measurements or within 24 hours after burn-in (or removal of bias) perform the final electrical parameter measurements.

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4.2.1 Percent defective allowable (PDA).

a. The PDA for class S devices shall be 5 percent for static burn-in and 5 percent for dynamic burn-in, based on the exact number of devices submitted to each separate burn-in.

b. Static burn-in I and II failure shall be cumulative for determining the PDA.

c. The PDA for class B devices shall be in accordance with MIL-PRF-38535 for static burn-in. Dynamic burn-in is not required.

d. Those devices whose measured characteristics, after burn-in, exceed the specified delta (Δ) limits or electrical parameter limits specified in table III, subgroup 1, are defective and shall be removed from the lot. The verified failures divided by the total number of devices in the lot initially submitted to burn-in shall be used to determine the percent defective for the lot and the lot shall be accepted or rejected based on the specified PDA.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-38535.

4.4 Technology Conformance inspection (TCI). Technology conformance inspection shall be in accordance with MIL-PRF-38535 and herein for groups A, B, C, D, and E inspections (see 4.4.1 through 4.4.5).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table III of MIL-PRF-38535 and as follows:

a. Tests shall be performed in accordance with table II herein.

b. Subgroups 5, 6, 7, and 8 shall be omitted.

c. Subgroup 4 (C_i measurement) shall be measured only for initial qualification and after process or design changes that may affect input capacitance. Capacitance shall be measured between the designated terminal and V_{SS} at a frequency of 1 MHz.

d. Subgroup 12 shall be added to the group A inspection requirements for class S devices using sample size series of 15 (acceptance 0) and consist of procedures, test conditions, and limits specified in table III.

e. Subgroups 9 and 11 shall be measured only for initial qualification and after process or design changes which may affect dynamic performance.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of MIL-PRF-38535.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table IV of MIL-PRF-38535 and as follows:

a. End-point electrical parameters shall be as specified in table II herein. Delta limits shall apply only to subgroup 1 of group C inspection and shall consist of tests specified in table IV herein.

b. The steady-state life test duration, test condition, and test temperature, or approved alternatives shall be as specified in the device manufacturer's QM plan in accordance with MIL-PRF-38535. The burn-in test circuit shall be maintained under document control by the device manufacturer's Technology Review Board (TRB) in accordance with MIL-PRF-38535 and shall be made available to the acquiring or preparing activity upon request. The test circuit shall specify the inputs, outputs, biases, and power dissipation, as applicable, in accordance with the intent specified in test method 1005 of MIL-STD-883.

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4.4.4 Group D inspection. Group D inspection shall be in accordance with table V of MIL-PRF-38535. End-point electrical parameters shall be as specified in table II herein.

4.4.5 Group E inspection. Group E inspection is required only for parts intended to be marked as radiation hardness assured (see 3.7 herein). RHA levels for device classes B and S shall be as specified in MIL-PRF-38535 and 4.5.4 herein.

4.5 Methods of inspection. Methods of inspection shall be specified and as follows:

4.5.1 Voltage and current. Unless otherwise specified, all voltages given are referenced to the microcircuit V_{SS} terminal. Currents given are conventional current and positive when flowing into the referenced terminal.

4.5.2 Burn-in and life test cool down procedures. When the burn-in and life tests are completed and prior to removal of bias voltages, the devices under test (DUT) shall be cooled to a temperature of 25°C ±3°C; then, electrical parameter end-point measurements shall be performed.

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TABLE III. Group A inspection for device type 01.

Symbol	MIL-STD-833 method	Case A,C,D,T,X,Y	Terminal conditions 1/														Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	1A	1B	1E	2E	2A	2B	V_{SS}	2C	2D	1I	2I	1C	1D	V_{PP}								
		Test No.																Min	Max	Min	Max	Min	Max	
V_{IC} (pos)		1	1 mA													GND	1A		1.5					Vdc
"		2		1 mA												"	1B		"					"
"		3												1 mA		"	1C		"					"
"		4													1 mA	"	1D		"					"
"		5										1mA				"	1I		"					"
"		6											1mA			"	2I		"					"
"		7					1 mA									"	2A		"					"
"		8						1 mA								"	2B		"					"
"		9								1 mA						"	2C		"					"
"		10									1 mA					"	2D		"					"
V_{IC} (neg)		11	-1 mA						GND								1A		-6.0					"
"		12		-1 mA					"								1B		"					"
"		13							"					-1 mA			1C		"					"
"		14							"						-1 mA		1D		"					"
"		15							"			-1mA					1I		"					"
"		16							"				-1mA				2I		"					"
"		17					-1 mA		"								2A		"					"
"		18						-1 mA	"								2B		"					"
"		19							"	-1 mA							2C		"					"
"		20							"		-1 mA						2D		"					"
I_{SS} 2/	3005	21	GND	GND			GND	GND	"	GND	GND	18.0V	18.0V	GND	GND	18.0V	V_{SS}		-25		-750			nA
"	"	22	GND	GND			GND	GND	"	GND	GND	GND	GND	GND	GND	"	"		"		"			"
"	"	23	18.0V	GND			18.0V	GND	"	18.0V	GND	"	"	18.0V	GND	"	"		"		"			"
"	"	24	GND	18.0V			GND	18.0V	"	GND	18.0V	"	"	GND	18.0V	"	"		"		"			"
"	"	25	GND	GND			GND	GND	"	18.0V	18.0V	"	"	18.0V	18.0V	"	"		"		"			"
"	"	26	18.0V	18.0V			18.0V	18.0V	"	GND	GND	"	"	GND	GND	"	"		"		"			"
"	"	27	18.0V	18.0V			18.0V	18.0V	"	18.0V	18.0V	18.0V	18.0V	18.0V	18.0V	"	"		"		"			"
V_{OL1}	3007	28	GND	GND			GND	GND	"	GND	GND	15.0V	GND	GND	GND	15.0V	1E		0.05		0.05		0.05	Vdc
"	"	29	15.0V	15.0V			"	"	"	"	"	GND	"	GND	GND	"	"		"		"		"	"
"	"	30	GND	GND			"	"	"	"	"	"	"	15.0V	15.0V	"	"		"		"		"	"
"	"	31	15.0V	15.0V			"	"	"	"	"	"	"	15.0V	15.0V	"	"		"		"		"	"
"	"	32	GND	GND			"	"	"	"	"	"	15.0V	GND	GND	"	2E		"		"		"	"
"	"	33	"	"			15.0V	15.0V	"	"	"	"	GND	"	"	"	"		"		"		"	"
"	"	34	"	"			GND	GND	"	15.0V	15.0V	"	"	"	"	"	"		"		"		"	"
"	"	35	"	"			15.0V	15.0V	"	15.0V	15.0V	"	"	"	"	"	"		"		"		"	"
V_{OH1}	3006	36	"	"			GND	GND	"	GND	GND	"	"	"	"	"	1E	14.95		14.95		14.95			"
"	"	37	"	"			"	"	"	"	"	"	"	"	"	"	2E	"		"		"			"
"	"	38	15.0V	"			"	"	"	"	"	"	"	15.0V	"	"	1E	"		"		"			"
"	"	39	GND	15.0V			"	"	"	"	"	"	"	GND	15.0V	"	1E	"		"		"			"
"	"	40	GND	GND			15.0V	"	"	15.0V	"	"	"	GND	GND	"	2E	"		"		"			"
"	"	41	GND	GND			GND	15.0V	"	GND	15.0V	"	"	GND	GND	"	2E	"		"		"			"

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 – Continued.

Symbol	MIL-STD-833 method	Case A,C,D,T,X,Y	Terminal conditions 1/														Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	1A	1B	1E	2E	2A	2B	V_{SS}	2C	2D	1I	2I	1C	1D	V_{DD}								
		Test No.																Min	Max	Min	Max	Min	Max	
V_{IL1}		42	1.5V	1.5V			1.5V	1.5V	GND	1.5V	1.5V	1.5V	1.5V	1.5V	1.5V	5.0V	1E	4.5		4.5		4.5		Vdc
V_{IH1}		43	3.5V	1.5V			"	"	"	"	"	"	"	3.5V	1.5V	"	"	4.5		4.5		4.5		"
"		44	1.5V	3.5V			"	"	"	"	"	"	"	1.5V	3.5V	"	"	4.5		4.5		4.5		"
"		45	3.5V	3.5V			"	"	"	"	"	"	"	1.5V	"	"	"		0.5		0.5		0.5	"
"		46	1.5V	1.5V			"	"	"	"	"	"	"	3.5V	"	"	"		"		"		"	"
"		47	3.5V	3.5V			"	"	"	"	"	"	"	3.5V	"	"	"		"		"		"	"
"		48	3.5V	3.5V			"	"	"	"	"	3.5V	"	3.5V	"	"	"		"		"		"	"
"		49	1.5V	1.5V			"	"	"	"	"	3.5V	"	1.5V	1.5V	"	"		"		"		"	"
"		50	"	"			"	"	"	"	"	1.5V	"	"	"	"	2E	4.5		4.5		4.5		"
"		51	"	"			3.5V	"	"	3.5V	"	"	"	"	"	"	"	4.5		4.5		4.5		"
"		52	"	"			1.5V	3.5V	"	1.5V	3.5V	"	"	"	"	"	"	4.5		4.5		4.5		"
"		53	"	"			3.5V	3.5V	"	1.5V	1.5V	"	"	"	"	"	"		0.5		0.5		0.5	"
"		54	"	"			1.5V	1.5V	"	3.5V	3.5V	"	"	"	"	"	"		"		"		"	"
"		55	"	"			3.5V	3.5V	"	3.5V	3.5V	"	"	"	"	"	"		"		"		"	"
"		56	"	"			3.5V	3.5V	"	3.5V	3.5V	"	3.5V	"	"	"	"		"		"		"	"
"		57	"	"			1.5V	1.5V	"	1.5V	1.5V	"	3.5V	"	"	"	"		"		"		"	"
V_{IL2}		58	3.0V	3.0V			3.0V	3.0V	"	3.0V	3.0V	3.0V	3.0V	3.0V	3.0V	10V	1E	9.0		9.0		9.0		"
V_{IH2}		59	7.0V	3.0V			"	"	"	"	"	"	"	7.0V	3.0V	"	"	9.0		9.0		9.0		"
"		60	3.0V	7.0V			"	"	"	"	"	"	"	3.0V	7.0V	"	"	9.0		9.0		9.0		"
"		61	7.0V	7.0V			"	"	"	"	"	"	"	3.0V	3.0V	"	"		1.0		1.0		1.0	"
"		62	3.0V	3.0V			"	"	"	"	"	"	"	7.0V	7.0V	"	"		"		"		"	"
"		63	7.0V	7.0V			"	"	"	"	"	"	"	7.0V	7.0V	"	"		"		"		"	"
"		64	7.0V	7.0V			"	"	"	"	"	7.0V	"	7.0V	7.0V	"	"		"		"		"	"
"		65	3.0V	3.0V			"	"	"	"	"	7.0V	"	3.0V	3.0V	"	"		"		"		"	"
"		66	"	"			"	"	"	3.0V	"	3.0V	"	"	"	"	2E	9.0		9.0		9.0		"
"		67	"	"			7.0V	"	"	7.0V	"	"	"	"	"	"	"	9.0		9.0		9.0		"
"		68	"	"			3.0V	7.0V	"	3.0V	7.0V	"	"	"	"	"	"	9.0		9.0		9.0		"
"		69	"	"			7.0V	"	"	3.0V	3.0V	"	"	"	"	"	"		1.0		1.0		1.0	"
"		70	"	"			3.0V	"	"	7.0V	7.0V	"	"	"	"	"	"		"		"		"	"
"		71	"	"			7.0V	"	"	7.0V	7.0V	"	"	"	"	"	"		"		"		"	"
"		72	"	"			7.0V	"	"	7.0V	7.0V	"	7.0V	"	"	"	"		"		"		"	"
"		73	"	"			3.0V	3.0V	"	3.0V	3.0V	"	7.0V	"	"	"	"		"		"		"	"
V_{IL3}		74	4.0V	4.0V			4.0V	4.0V	"	4.0V	4.0V	4.0V	4.0V	4.0V	4.0V	15V	1E	13.5		13.5		13.5		"
V_{IH3}		75	11.0V	4.0V			"	"	"	"	"	"	"	11.0V	4.0V	"	"	13.5		13.5		13.5		"
"		76	4.0V	11.0V			"	"	"	"	"	"	"	4.0V	11.0V	"	"	13.5		13.5		13.5		"
"		77	11.0V	11.0V			"	"	"	"	"	"	"	4.0V	4.0V	"	"		1.5		1.5		1.5	"
"		78	4.0V	4.0V			"	"	"	"	"	"	"	11.0V	11.0V	"	"		"		"		"	"
"		79	11.0V	11.0V			"	"	"	"	"	"	"	11.0V	11.0V	"	"		"		"		"	"
"		80	11.0V	11.0V			"	"	"	"	"	11.0V	"	11.0V	11.0V	"	"		"		"		"	"
"		81	4.0V	4.0V			"	"	"	"	"	11.0V	"	4.0V	4.0V	"	"		"		"		"	"
"		82	"	"			"	"	"	"	"	4.0V	"	"	"	"	2E	13.5		13.5		13.5		"
"		83	"	"			11.0V	"	"	11.0V	"	"	"	"	"	"	"	13.5		13.5		13.5		"
"		84	"	"			4.0V	11.0V	"	4.0V	11.0V	"	"	"	"	"	"	13.5		13.5		13.5		"
"		85	"	"			11.0V	11.0V	"	4.0V	4.0V	"	"	"	"	"	"		1.5		1.5		1.5	"
"		86	"	"			4.0V	4.0V	"	11.0V	11.0V	"	"	"	"	"	"		"		"		"	"
"		87	"	"			11.0V	11.0V	"	11.0V	11.0V	"	"	"	"	"	"		"		"		"	"
"		88	"	"			11.0V	11.0V	"	11.0V	11.0V	"	11.0V	"	"	"	"		"		"		"	"
"		89	"	"			4.0V	4.0V	"	4.0V	4.0V	"	11.0V	"	"	"	"		"		"		"	"

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TABLE III. Group A inspection for device type 01 – Continued.

Symbol	MIL-STD-833 method	Case A,C,D,T,X,Y	Terminal conditions 1/														Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	1A	1B	1E	2E	2A	2B	V_{SS}	2C	2D	1I	2I	1C	1D	V_{DD}								
		Test No.																Min	Max	Min	Max	Min	Max	
3/ I_{IH1}	3010	90	18.0V	18.0V			18.0V	18.0V	GND	18.0V	18.0V	18.0V	18.0V	18.0V	18.0V	18.0V	All		10					nA
I_{IH2}	"	91	GND	GND			GND	GND	"	GND	GND	18.0V	GND	GND	GND	"	1I		1		45			"
"	"	92	18.0V	GND			"	"	"	"	"	GND	"	GND	"	"	1A		"		"			"
"	"	93	GND	18.0V			"	"	"	"	"	"	"	GND	"	"	1B		"		"			"
"	"	94	"	GND			"	"	"	"	"	"	"	18.0V	"	"	1C		"		"			"
"	"	95	"	"			"	"	"	"	"	"	"	GND	18.0V	"	1D		"		"			"
"	"	96	"	"			"	"	"	"	"	"	18.0V	"	GND	"	2I		"		"			"
"	"	97	"	"			18.0V	"	"	"	"	"	GND	"	"	"	2A		"		"			"
"	"	98	"	"			GND	18.0V	"	"	"	"	"	"	"	"	2B		"		"			"
"	"	99	"	"			"	GND	"	18.0V	"	"	"	"	"	"	2C		"		"			"
"	"	100	"	"			"	GND	"	GND	18.0V	"	"	"	"	"	2D		"		"			"
3/ I_{IL2}	3009	101	"	"			"	GND	"	GND	GND	"	"	"	"	"	All		-10					"
I_{IL2}	"	102	18.0V	18.0V			18.0V	18.0V	"	18.0V	18.0V	"	18.0V	18.0V	18.0V	"	1I		-1		-45			"
"	"	103	GND	18.0V			"	"	"	"	"	18.0V	"	18.0V	"	"	1A		"		"			"
"	"	104	"	GND			"	"	"	"	"	"	"	18.0V	"	"	1B		"		"			"
"	"	105	"	18.0V			"	"	"	"	"	"	"	GND	"	"	1C		"		"			"
"	"	106	"	"			"	"	"	"	"	"	"	18.0V	GND	"	1D		"		"			"
"	"	107	"	"			"	"	"	"	"	"	GND	"	18.0V	"	2I		"		"			"
"	"	108	"	"			GND	"	"	"	"	"	18.0V	"	"	"	2A		"		"			"
"	"	109	"	"			18.0V	GND	"	"	"	"	"	"	"	"	2B		"		"			"
"	"	110	"	"			18.0V	18.0V	"	GND	"	"	"	"	"	"	2C		"		"			"
"	"	111	"	"			18.0V	18.0V	"	18.0V	GND	"	"	"	"	"	2D		"		"			"
I_{OL1}		112	GND	GND	0.4V		GND	GND	"	GND	"	5.0V	GND	GND	GND	5.0V	1E	0.51		0.36		0.64		mA
I_{OL1}		113	"	"		0.4V	"	"	"	"	"	GND	5.0V	"	"	5.0V	2E	0.51		0.36		0.64		"
I_{OL2}		114	"	"	1.5V		"	"	"	"	"	15.0V	GND	"	"	15.0V	1E	3.4		2.4		4.2		"
I_{OL2}		115	"	"		1.5V	"	"	"	"	"	GND	15.0V	"	"	15.0V	2E	3.4		2.4		4.2		"
I_{OH1}		116	"	"	4.6V		"	"	"	"	"	"	GND	"	"	5.0V	1E	-0.51		-0.36		-0.64		"
I_{OH1}		117	"	"		4.6V	"	"	"	"	"	"	"	"	"	5.0V	2E	-0.51		-0.36		-0.64		"
I_{OH2}		118	"	"	13.5V		"	"	"	"	"	"	"	"	"	15.0V	1E	-3.4		-2.4		-4.2		"
I_{OH2}		119	"	"		13.5V	"	"	"	"	"	"	"	"	"	15.0V	2E	-3.4		-2.4		-4.2		"
																		Subgroup 4 T_A = 25°C						
																		Min	Max					
C_i	3012	120							GND			4/				GND	1I		7.5					pF
"	"	121	4/						"							"	1A		"					"
"	"	122		4/					"							"	1B		"					"
"	"	123							"					4/		"	1C		"					"
"	"	124							"						4/	"	1D		"					"
"	"	125							"				4/			"	2I		"					"
"	"	126					4/		"							"	2A		"					"
"	"	127						4/	"							"	2B		"					"
"	"	128							"					4/		"	1C		"					"
"	"	129A							"						4/	"	1D		"					"
"	"	129B							"	4/						"	2C		"					"
"	"	129C							"		4/					"	2D		"					"

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TABLE III. Group A inspection for device type 01 – Continued.

Symbol	MIL-STD-833 method	Case A,C,D,T,X,Y	Terminal conditions 1/														Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
		Symbol	1A	1B	1E	2E	2A	2B	V_{SS}	2C	2D	1I	2I	1C	1D	V_{DD}								
		Test No.																Min	Max	Min	Max	Min	Max	
t_{PHL}	3003	130	GND	GND	OUT		GND	GND	GND	GND	GND	IN	GND	GND	GND	5.0V	1I to 1E	15	300	21	420	15	300	ns
"	Fig. 3	131	IN	5.0V	"		"	"	"	"	"	GND	"	GND	GND	"	1A to 1E	23	450	32	630	23	450	"
"	"	132	5.0V	IN	"		"	"	"	"	"	"	"	GND	GND	"	1B to 1E	"	"	"	"	"	"	"
"	"	133	GND	GND	"		"	"	"	"	"	"	"	IN	5.0V	"	1C to 1E	"	"	"	"	"	"	"
"	"	134	"	"	"		"	"	"	"	"	"	"	5.0V	IN	"	1D to 1E	"	"	"	"	"	"	"
"	"	135	"	"		OUT	"	"	"	"	"	"	IN	GND	GND	"	2I to 2E	15	300	21	420	15	300	"
"	"	136	"	"		"	IN	5.0V	"	"	"	"	GND	"	"	"	2A to 2E	23	450	32	630	23	450	"
"	"	137	"	"		"	5.0V	IN	"	"	"	"	"	"	"	"	2B to 2E	"	"	"	"	"	"	"
"	"	138	"	"		"	GND	GND	"	IN	5.0V	"	"	"	"	"	2C to 2E	"	"	"	"	"	"	"
"	"	139	"	"		"	"	"	"	5.0V	IN	"	"	"	"	"	2D to 2E	"	"	"	"	"	"	"
t_{PLH}	"	140	"	"	OUT		"	"	"	GND	GND	IN	"	"	"	"	1I to 1E	25	500	35	700	25	500	"
"	"	141	IN	5.0V	"		"	"	"	"	"	GND	"	"	"	"	1A to 1E	31	620	44	870	31	620	"
"	"	142	5.0V	IN	"		"	"	"	"	"	"	"	"	"	"	1B to 1E	"	"	"	"	"	"	"
"	"	143	GND	GND	"		"	"	"	"	"	"	"	IN	5.0V	"	1C to 1E	"	"	"	"	"	"	"
"	"	144	"	"	"		"	"	"	"	"	"	"	5.0V	IN	"	1D to 1E	"	"	"	"	"	"	"
"	"	145	"	"		OUT	"	"	"	"	"	"	IN	GND	GND	"	2I to 2E	25	500	35	700	25	500	"
"	"	146	"	"		"	IN	5.0V	"	"	"	"	GND	"	"	"	2A to 2E	31	620	44	870	31	620	"
"	"	147	"	"		"	5.0V	IN	"	"	"	"	"	"	"	"	2B to 2E	"	"	"	"	"	"	"
"	"	148	"	"		"	GND	GND	"	IN	5.0V	"	"	"	"	"	2C to 2E	"	"	"	"	"	"	"
"	"	149	"	"		"	"	"	"	5.0V	IN	"	"	"	"	"	2D to 2E	"	"	"	"	"	"	"
t_{THL}	3004	150	"	"	OUT		"	"	"	GND	GND	IN	"	"	"	"	1E	10	200	14	200	10	200	"
"	Fig. 3	151	IN	5.0V	"		"	"	"	"	"	GND	"	"	"	"	"	"	"	"	"	"	"	"	"
"	"	152	5.0V	IN	"		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
"	"	153	GND	GND	"		"	"	"	"	"	"	"	IN	5.0V	"	"	"	"	"	"	"	"	"	"
"	"	154	"	"	"		"	"	"	"	"	"	"	5.0V	IN	"	"	"	"	"	"	"	"	"	"
"	"	155	"	"		OUT	"	"	"	"	"	"	IN	GND	GND	"	2E	"	"	"	"	"	"	"
"	"	156	"	"		"	IN	5.0V	"	"	"	"	GND	"	"	"	"	"	"	"	"	"	"	"
"	"	157	"	"		"	5.0V	IN	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
"	"	158	"	"		"	GND	GND	"	IN	5.0V	"	"	"	"	"	"	"	"	"	"	"	"	"	"
"	"	159	"	"		"	"	"	"	5.0V	IN	"	"	"	"	"	"	"	"	"	"	"	"	"	"
t_{TLH}	"	160	"	"	OUT		"	"	"	GND	GND	IN	"	"	"	"	1E	"	"	"	"	"	"	"
"	"	161	IN	5.0V	"		"	"	"	"	"	GND	"	"	"	"	"	"	"	"	"	"	"	"	"
"	"	162	5.0V	IN	"		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
"	"	163	GND	GND	"		"	"	"	"	"	"	"	IN	5.0V	"	"	"	"	"	"	"	"	"	"
"	"	164	"	"	"		"	"	"	"	"	"	"	5.0V	IN	"	"	"	"	"	"	"	"	"	"
"	"	165	"	"		OUT	"	"	"	"	"	"	IN	GND	GND	"	2E	"	"	"	"	"	"	"
"	"	166	"	"		"	IN	5.0V	"	"	"	"	GND	"	"	"	"	"	"	"	"	"	"	"	"
"	"	167	"	"		"	5.0V	IN	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
"	"	168	"	"		"	GND	GND	"	IN	5.0V	"	"	"	"	"	"	"	"	"	"	"	"	"	"
"	"	169	"	"		"	GND	GND	"	5.0V	IN	"	"	"	"	"	"	"	"	"	"	"	"	"	"
																		Subgroup 12 T_A = 25°C						
t_{PHL}	3003	170	GND	GND	OUT		GND	GND	GND	GND	GND	IN	GND	GND	GND	10.0V	1I to 1E	6	120					
"	Fig. 3	171	IN	10.0V	"		"	"	"	"	"	GND	"	GND	GND	"	1A to 1E	9	180					
"	"	172	10.0V	IN	"		"	"	"	"	"	"	"	GND	GND	"	1B to 1E	"	"					
"	"	173	GND	GND	"		"	"	"	"	"	"	"	IN	10.0V	"	1C to 1E	"	"					
"	"	174	"	"	"		"	"	"	"	"	"	"	10.0V	IN	"	1D to 1E	"	"					
"	"	175	"	"		OUT	"	"	"	"	"	"	IN	GND	GND	"	2I to 2E	6	120					
"	"	176	"	"		"	IN	10.0V	"	"	"	"	GND	"	"	"	2A to 2E	9	180					
"	"	177	"	"		"	10.0V	IN	"	"	"	"	"	"	"	"	2B to 2E	"	"					
"	"	178	"	"		"	GND	GND	"	IN	10.0V	"	"	"	"	"	2C to 2E	"	"					
"	"	179	"	"		"	GND	GND	"	10.0V	IN	"	"	"	"	"	2D to 2E	"	"					

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 – Continued.

Symbol	MIL-STD-833 method	Case A,C,D,T,X,Y	Terminal conditions 1/														Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 12 T_A = 25°C						
		Symbol	1A	1B	1E	2E	2A	2B	V_{SS}	2C	2D	1I	2I	1C	1D	V_{DD}								
		Test No.																Min	Max	Min	Max	Min	Max	
t_{PLH}	3003	180	GND	GND	OUT		GND	GND	GND	GND	GND	IN	GND	GND	GND	10.0V	1I to 1E	10	200					ns
"	Fig. 3	181	IN	10.0V	"		"	"	"	"	"	GND	"	GND	GND	"	1A to 1E	13	250					"
"	"	182	10.0V	IN	"		"	"	"	"	"	"	"	GND	GND	"	1B to 1E	"	"					"
"	"	183	GND	GND	"		"	"	"	"	"	"	"	IN	10.0V	"	1C to 1E	"	"					"
"	"	184	"	"	"		"	"	"	"	"	"	"	10.0V	IN	"	1D to 1E	"	"					"
"	"	185	"	"		OUT	"	"	"	"	"	"	IN	GND	GND	"	2I to 2E	10	200					"
"	"	186	"	"		"	IN	10.0V	"	"	"	"	GND	"	"	"	2A to 2E	13	250					"
"	"	187	"	"		"	10.0V	IN	"	"	"	"	"	"	"	"	2B to 2E	"	"					"
"	"	188	"	"		"	GND	GND	"	IN	10.0V	"	"	"	"	"	2C to 2E	"	"					"
"	"	189	"	"		"	"	"	"	10.0V	IN	"	"	"	"	"	2D to 2E	"	"					"
t_{THL}	3004	190	"	"	OUT		"	"	"	GND	GND	IN	"	"	"	"	1E	5	100					"
"	Fig. 3	191	IN	10.0V	"		"	"	"	"	"	GND	"	"	"	"	"	"	"					"
"	"	192	10.0V	IN	"		"	"	"	"	"	"	"	"	"	"	"	"	"					"
"	"	193	GND	GND	"		"	"	"	"	"	"	"	IN	10.0V	"	"	"	"					"
"	"	194	10.0V	"	"		"	"	"	"	"	"	"	10.0V	IN	"	"	"	"					"
"	"	195	"	"		OUT	"	"	"	"	"	"	IN	GND	GND	"	2E	"	"					"
"	"	196	"	"		"	IN	10.0V	"	"	"	"	GND	"	"	"	"	"	"					"
"	"	197	"	"		"	10.0V	IN	"	"	"	"	"	"	"	"	"	"	"					"
"	"	198	"	"		"	GND	GND	"	IN	10.0V	"	"	"	"	"	"	"	"					"
"	"	199	"	"		"	"	"	"	10.0V	IN	"	"	"	"	"	"	"	"					"
t_{TLH}	"	200	"	"	OUT		"	"	"	GND	GND	IN	"	"	"	"	1E	"	"					"
"	"	201	IN	10.0V	"		"	"	"	"	"	GND	"	"	"	"	"	"	"					"
"	"	202	10.0V	IN	"		"	"	"	"	"	"	"	"	"	"	"	"	"					"
"	"	203	GND	GND	"		"	"	"	"	"	"	"	IN	10.0V	"	"	"	"					"
"	"	204	"	"	"		"	"	"	"	"	"	"	10.0V	IN	"	"	"	"					"
"	"	205	"	"		OUT	"	"	"	"	"	"	IN	GND	GND	"	2E	"	"					"
"	"	206	"	"		"	IN	10.0V	"	"	"	"	GND	"	"	"	"	"	"					"
"	"	207	"	"		"	10.0V	IN	"	"	"	"	"	"	"	"	"	"	"					"
"	"	208	"	"		"	GND	GND	"	IN	10.0V	"	"	"	"	"	"	"	"					"
"	"	209	"	"		"	GND	GND	"	10.0V	IN	"	"	"	"	"	"	"	"					"

1/ Pins not designated may be "HIGH" level logic, "LOW" level logic or open. Exceptions are as follows: $V_{IC(pos)}$ tests, the V_{SS} terminals shall be open; $V_{IC(neg)}$ tests, the V_{DD} terminal shall be open; I_{SS} tests, the output terminals shall be open.
2/ The I_{SS} measurements shall be performed in sequence.
3/ The device manufacturer may, at his option, measure I_{IL} and I_{IH} at 25°C for each individual input or measure all inputs together.
4/ See 4.4.1c.

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TABLE III. Group A inspection for device type 02.

Symbol	MIL-STD-833 method	Case A,C,D,T,X,Y	Terminal conditions 1/														Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	A	B	J	NC	E	F	V_{SS}	G	H	I	EN	C	D	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																						
V_{IC} (pos)		1	1mA													GND	A		1.5					Vdc
"		2		1mA												"	B		"					"
"		3												1mA		"	C		"					"
"		4													1mA	"	D		"					"
"		5					1mA									"	E		"					"
"		6						1mA								"	F		"					"
"		7								1mA						"	G		"					"
"		8									1mA					"	H		"					"
"		9										1mA				"	I		"					"
"		10											1mA			"	EN		"					"
V_{IC} (neg)		11	-1mA						GND								A		-6.0					"
"		12		-1mA					"								B		"					"
"		13							"					-1mA			C		"					"
"		14							"						-1mA		D		"					"
"		15					-1mA		"								E		"					"
"		16						-1mA	"								F		"					"
"		17							"	-1mA							G		"					"
"		18							"		-1mA						H		"					"
"		19							"			-1mA					I		"					"
"		20							"				-1mA				EN		"					"
I_{SS} 2/	3005	21	GND	GND			GND	GND	"	GND	GND	GND	18.0V	GND	GND	18.0V	V_{SS}		-25		-750			nA
"	"	22	GND	GND			GND	GND	"	GND	GND	18.0V	18.0V	GND	GND	"	"		"		"				"
"	"	23	GND	GND			GND	GND	"	GND	GND	GND	GND	GND	GND	"	"		"		"				"
"	"	24	18.0V	18.0V			18.0V	18.0V	"	18.0V	18.0V	18.0V	18.0V	18.0V	18.0V	"	"		"		"				"
"	"	25	18.0V	18.0V			GND	GND	"	GND	GND	GND	"	GND	GND	"	"		"		"				"
"	"	26	GND	GND			GND	GND	"	GND	GND	"	"	18.0V	18.0V	"	"		"		"				"
"	"	27	GND	"			18.0V	18.0V	"	GND	GND	"	"	GND	GND	"	"		"		"				"
"	"	28	GND	"			GND	GND	"	18.0V	18.0V	"	"	GND	"	"	"		"		"				"
"	"	29	18.0V	"			18.0V	"	"	18.0V	GND	"	"	18.0V	"	"	"		"		"				"
V_{OL1}	3007	30	GND	"			GND	"	"	GND	GND	15.0V	15.0V	GND	"	15.0V	J		0.05		0.05		0.05	Vdc
"	"	31	GND	"			GND	"	"	GND	GND	GND	GND	GND	"	"	"		"		"		"	"
"	"	32	15.0V	15.0V			15.0V	15.0V	"	15.0V	15.0V	"	15.0V	15.0V	15.0V	"	"		"		"		"	"
"	"	33	15.0V	15.0V			GND	GND	"	GND	GND	"	"	GND	GND	"	"		"		"		"	"
"	"	34	GND	GND			GND	GND	"	GND	GND	"	"	15.0V	15.0V	"	"		"		"		"	"
"	"	35	"	"			15.0V	15.0V	"	15.0V	15.0V	"	"	GND	GND	"	"		"		"		"	"
"	"	36	"	"			GND	GND	"	15.0V	15.0V	"	"	GND	"	"	"		"		"		"	"
V_{OH1}	3006	37	"	"			GND	GND	"	GND	15.0V	"	"	GND	"	"	"	14.95		14.95		14.95		"
V_{OH1}	3006	38	15.0V	"			15.0V	GND	"	15.0V	GND	"	"	15.0V	"	"	"	14.95		14.95		14.95		"
V_{OH1}	3006	39	GND	15.0V			GND	15.0V	"	GND	15.0V	"	"	GND	15.0V	"	"	14.95		14.95		14.95		"

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-833 method	Case A,C,D,T,X,Y	Terminal conditions 1/														Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	A	B	J	NC	E	F	V_{SS}	G	H	I	EN	C	D	V_{DD}								
		Test No.																Min	Max	Min	Max	Min	Max	
V_{IL1}		40	1.5V	1.5V			1.5V	1.5V	GND	1.5V	1.5V	1.5V	3.5V	1.5V	1.5V	5.0V	J	4.5		4.5		4.5		Vdc
V_{IH1}		41	1.5V	1.5V			1.5V	1.5V	"	1.5V	1.5V	3.5V	3.5V	1.5V	1.5V	"	"		0.5		0.5		0.5	"
"		42	1.5V	1.5V			1.5V	1.5V	"	1.5V	1.5V	1.5V	1.5V	1.5V	1.5V	"	"		"		"		"	"
"		43	3.5V	3.5V			3.5V	3.5V	"	3.5V	3.5V	3.5V	3.5V	3.5V	3.5V	"	"		"		"		"	"
"		44	3.5V	3.5V			1.5V	1.5V	"	1.5V	1.5V	1.5V	"	1.5V	1.5V	"	"		"		"		"	"
"		45	1.5V	1.5V			3.5V	3.5V	"	1.5V	1.5V	"	"	1.5V	1.5V	"	"		"		"		"	"
"		46	1.5V	"			1.5V	1.5V	"	3.5V	3.5V	"	"	1.5V	1.5V	"	"		"		"		"	"
"		47	1.5V	"			1.5V	1.5V	"	1.5V	1.5V	"	"	3.5V	3.5V	"	"		"		"		"	"
"		48	3.5V	"			3.5V	1.5V	"	3.5V	1.5V	"	"	3.5V	1.5V	"	"	4.5		4.5		4.5		"
"		49	1.5V	3.5V			1.5V	3.5V	"	1.5V	3.5V	"	"	1.5V	3.5V	"	"	4.5		4.5		4.5		"
V_{IL2}		50	3.0V	3.0V			3.0V	3.0V	"	3.0V	3.0V	3.0V	7.0V	3.0V	3.0V	10.0V	"	9.0		9.0		9.0		"
V_{IH2}		51	3.0V	3.0V			3.0V	3.0V	"	3.0V	3.0V	7.0V	7.0V	3.0V	3.0V	"	"		1.0		1.0		1.0	"
"		52	3.0V	3.0V			3.0V	3.0V	"	3.0V	3.0V	3.0V	3.0V	3.0V	3.0V	"	"		"		"		"	"
"		53	7.0V	7.0V			7.0V	7.0V	"	7.0V	7.0V	7.0V	7.0V	7.0V	7.0V	"	"		"		"		"	"
"		54	7.0V	7.0V			3.0V	3.0V	"	3.0V	3.0V	3.0V	"	3.0V	3.0V	"	"		"		"		"	"
"		55	3.0V	3.0V			7.0V	7.0V	"	3.0V	3.0V	"	"	3.0V	3.0V	"	"		"		"		"	"
"		56	3.0V	"			3.0V	3.0V	"	7.0V	7.0V	"	"	3.0V	3.0V	"	"		"		"		"	"
"		57	3.0V	"			3.0V	3.0V	"	3.0V	3.0V	"	"	7.0V	7.0V	"	"		"		"		"	"
"		58	7.0V	"			7.0V	3.0V	"	7.0V	3.0V	"	"	7.0V	3.0V	"	"	9.0		9.0		9.0		"
"		59	3.0V	7.0V			3.0V	7.0V	"	3.0V	7.0V	"	"	3.0V	7.0V	"	"	9.0		9.0		9.0		"
V_{IL3}		60	4.0V	4.0V			4.0V	4.0V	"	4.0V	4.0V	4.0V	11.0V	4.0V	4.0V	15.0V	"	13.5		13.5		13.5		"
V_{IH3}		61	4.0V	4.0V			4.0V	4.0V	"	4.0V	4.0V	11.0V	11.0V	4.0V	4.0V	"	"		1.5		1.5		1.5	"
"		62	4.0V	4.0V			4.0V	4.0V	"	4.0V	4.0V	4.0V	4.0V	4.0V	4.0V	"	"		"		"		"	"
"		63	11.0V	11.0V			11.0V	11.0V	"	11.0V	11.0V	11.0V	11.0V	11.0V	11.0V	"	"		"		"		"	"
"		64	11.0V	11.0V			4.0V	4.0V	"	4.0V	4.0V	4.0V	"	4.0V	4.0V	"	"		"		"		"	"
"		65	4.0V	4.0V			11.0V	11.0V	"	4.0V	4.0V	"	"	4.0V	4.0V	"	"		"		"		"	"
"		66	4.0V	"			4.0V	4.0V	"	11.0V	11.0V	"	"	4.0V	4.0V	"	"		"		"		"	"
"		67	4.0V	"			4.0V	4.0V	"	4.0V	4.0V	"	"	11.0V	11.0V	"	"		"		"		"	"
"		68	11.0V	"			11.0V	4.0V	"	11.0V	4.0V	"	"	11.0V	4.0V	"	"	13.5		13.5		13.5		"
"		69	4.0V	11.0V			4.0V	11.0V	"	4.0V	11.0V	"	"	4.0V	11.0V	"	"	13.5		13.5		13.5		"
I_{IH1} 3/	3010	70	18.0V	18.0V			18.0V	18.0V	"	18.0V	18.0V	18.0V	18.0V	18.0V	18.0V	18.0V	All inputs together		10					nA
I_{IH2}	3010	71	18.0V	GND			GND	GND	"	GND	GND	GND	GND	GND	GND	"	A		1		45			"
"	"	72	GND	18.0V			"	"	"	"	"	"	"	GND	GND	"	B		"		"			"
"	"	73	"	GND			"	"	"	"	"	"	"	18.0V	GND	"	C		"		"			"
"	"	74	"	"			"	"	"	"	"	"	"	GND	18.0V	"	D		"		"			"
"	"	75	"	"			18.0V	"	"	"	"	"	"	"	GND	"	E		"		"			"
"	"	76	"	"			GND	18.0V	"	"	"	"	"	"	"	"	F		"		"			"
"	"	77	"	"			"	GND	"	18.0V	"	"	"	"	"	"	G		"		"			"
"	"	78	"	"			"	"	"	GND	18.0V	"	"	"	"	"	H		"		"			"
"	"	79	"	"			"	"	"	GND	GND	18.0V	"	"	"	"	I		"		"			"
"	"	80	"	"			"	"	"	GND	GND	GND	18.0V	"	"	"	EN		"		"			"

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 – Continued.

Symbol	MIL-STD-833 method	Case A,C,D,T,X,Y	Terminal conditions 1/														Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	A	B	J	NC	E	F	V_{SS}	G	H	I	EN	C	D	V_{DD}								
		Test No.																Min	Max	Min	Max	Min	Max	
I_{IL1} 3/	3009	81	GND	GND			GND	GND	GND	GND	GND	GND	GND	GND	GND	18.0V	All inputs together		-10					nA
I_{IL2}	"	82	GND	18.0V			18.0V	18.0V	"	18.0V	18.0V	18.0V	18.0V	18.0V	18.0V	"	A		-1		-45			"
"	"	83	18.0V	GND			"	"	"	"	"	"	"	18.0V	18.0V	"	B		"		"			"
"	"	84	"	18.0V			"	"	"	"	"	"	"	GND	18.0V	"	C		"		"			"
"	"	85	"	"			"	"	"	"	"	"	"	18.0V	GND	"	D		"		"			"
"	"	86	"	"			GND	"	"	"	"	"	"	"	18.0V	"	E		"		"			"
"	"	87	"	"			18.0V	GND	"	"	"	"	"	"	"	"	F		"		"			"
"	"	88	"	"			"	18.0V	"	GND	"	"	"	"	"	"	G		"		"			"
"	"	89	"	"			"	"	"	18.0V	GND	"	"	"	"	"	H		"		"			"
"	"	90	"	"			"	"	"	18.0V	18.0V	GND	"	"	"	"	I		"		"			"
"	"	91	"	"			"	"	"	18.0V	18.0V	18.0V	GND	"	"		EN		"		"			"
I_{OL1}		92	GND	GND	0.4V		GND	GND	"	GND	GND	GND	GND	GND	GND	5.0V	J	0.51		0.36		0.64		mA
I_{OL2}		93	"	"	1.5V		"	"	"	"	"	"	GND	"	"	15.0V	"	3.4		2.4		4.2		"
I_{OH1}		94	"	"	4.6V		"	"	"	"	"	"	5.0V	"	"	5.0V	"	-0.51		-0.36		-0.64		"
I_{OH2}		95	"	"	13.5V		"	"	"	"	"	"	15.0V	"	"	15.0V	"	-3.4		-2.4		-4.2		"
																		Subgroup 4 T_A = 25°C						
C_i	3012	96	4/						GND							GND	A		7.5					pF
"	"	97		4/					"							"	B		"					"
"	"	98							"					4/		"	C		"					"
"	"	99							"						4/	"	D		"					"
"	"	100					4/		"							"	E		"					"
"	"	101						4/	"							"	F		"					"
"	"	102							"	4/						"	G		"					"
"	"	103							"		4/					"	H		"					"
"	"	104							"			4/				"	I		"					"
"	"	105							"				4/			"	EN		"					"
																		Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		Unit
t_{PHL}	3003	106	GND	GND	OUT		GND	GND	GND	GND	GND	IN	5.0V	GND	GND	5.0V	I to J	15	300	21	420	15	300	ns
"	Fig. 3	107	IN	5.0V	"		"	"	"	"	"	GND	"	GND	GND	"	A to J	23	450	32	630	23	450	"
"	"	108	5.0V	IN	"		"	"	"	"	"	"	"	GND	GND	"	B to J	"	"	"	"	"	"	"
"	"	109	GND	GND	"		"	"	"	"	"	"	"	IN	5.0V	"	C to J	"	"	"	"	"	"	"
"	"	110	"	"	"		"	"	"	"	"	"	"	5.0V	IN	"	D to J	"	"	"	"	"	"	"
"	"	111	"	"	"		IN	5.0V	"	"	"	"	"	GND	GND	"	E to J	"	"	"	"	"	"	"
"	"	112	"	"	"		5.0V	IN	"	"	"	"	"	"	"	"	F to J	"	"	"	"	"	"	"
"	"	113	"	"	"		GND	GND	"	IN	5.0V	"	"	"	"	"	G to J	"	"	"	"	"	"	"
"	"	114	"	"	"		"	"	"	5.0V	IN	"	"	"	"	"	H to J	"	"	"	"	"	"	"
"	"	115	"	"	"		"	"	"	GND	GND	"	IN	"	"	"	EN to J	"	"	"	"	"	"	"
t_{PLH}	"	116	"	"	"		"	"	"	"	"	IN	5.0V	"	"	"	I to J	25	500	35	700	25	500	"
"	"	117	IN	5.0V	"		"	"	"	"	"	GND	"	"	"	"	A to J	35	700	49	980	35	700	"
"	"	118	5.0V	IN	"		"	"	"	"	"	"	"	"	"	"	B to J	"	"	"	"	"	"	"
"	"	119	GND	GND	"		"	"	"	"	"	"	"	IN	5.0V	"	C to J	"	"	"	"	"	"	"
"	"	120	"	"	"		"	"	"	"	"	"	"	5.0V	IN	"	D to J	"	"	"	"	"	"	"
"	"	121	"	"	"		IN	5.0V	"	"	"	"	"	GND	GND	"	E to J	"	"	"	"	"	"	"
"	"	122	"	"	"		5.0V	IN	"	"	"	"	"	"	"	"	F to J	"	"	"	"	"	"	"
"	"	123	"	"	"		GND	GND	"	IN	5.0V	"	"	"	"	"	G to J	"	"	"	"	"	"	"
"	"	124	"	"	"		GND	GND	"	5.0V	IN	"	"	"	"	"	H to J	"	"	"	"	"	"	"
"	"	125	"	"	"		GND	GND	"	GND	GND	"	IN	"	"	"	EN to J	"	"	"	"	"	"	"

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 – Continued.

Symbol	MIL-STD-833 method	Case A,C,D,T,X,Y	Terminal conditions 1/														Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
		Symbol	A	B	J	NC	E	F	V_{SS}	G	H	I	EN	C	D	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																						
t_{THL}	3004	126	GND	GND	OUT		GND	GND	GND	GND	GND	IN	5.0V	GND	GND	5.0V	J	10	200	18	280	10	200	ns
"	Fig. 3	127	IN	5.0V	"		"	"	"	"	"	GND	"	GND	GND	"	"	"	"	"	"	"	"	"
"	"	128	5.0V	IN	"		"	"	"	"	"	"	"	GND	GND	"	"	"	"	"	"	"	"	"
"	"	129	GND	GND	"		"	"	"	"	"	"	"	IN	5.0V	"	"	"	"	"	"	"	"	"
"	"	130	"	"	"		"	"	"	"	"	"	"	5.0V	IN	"	"	"	"	"	"	"	"	"
"	"	131	"	"	"		IN	5.0V	"	"	"	"	"	GND	GND	"	"	"	"	"	"	"	"	"
"	"	132	"	"	"		5.0V	IN	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
"	"	133	"	"	"		GND	GND	"	IN	5.0V	"	"	"	"	"	"	"	"	"	"	"	"	"
"	"	134	"	"	"		"	"	"	5.0V	IN	"	"	"	"	"	"	"	"	"	"	"	"	"
"	"	135	"	"	"		"	"	"	GND	GND	"	IN	"	"	"	"	"	"	"	"	"	"	"
t_{TLH}	"	136	"	"	"		"	"	"	"	"	IN	5.0V	"	"	"	"	"	"	"	"	"	"	"
"	"	137	IN	5.0V	"		"	"	"	"	"	GND	"	"	"	"	"	"	"	"	"	"	"	"
"	"	138	5.0V	IN	"		"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
"	"	139	GND	GND	"		"	"	"	"	"	"	"	IN	5.0V	"	"	"	"	"	"	"	"	"
"	"	140	"	"	"		"	"	"	"	"	"	"	5.0V	IN	"	"	"	"	"	"	"	"	"
"	"	141	"	"	"		IN	5.0V	"	"	"	"	"	GND	GND	"	"	"	"	"	"	"	"	"
"	"	142	"	"	"		5.0V	IN	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"
"	"	143	"	"	"		GND	GND	"	IN	5.0V	"	"	"	"	"	"	"	"	"	"	"	"	"
"	"	144	"	"	"		GND	GND	"	5.0V	IN	"	"	"	"	"	"	"	"	"	"	"	"	"
"	"	145	"	"	"		GND	GND	"	GND	GND	"	IN	"	"	"	"	"	"	"	"	"	"	"
																		Subgroup 12 T_A = 25°C						
t_{PHL}	3003	146	GND	GND	OUT		GND	GND	GND	GND	GND	IN	10.0V	GND	GND	10.0V	I to J	6	120					ns
"	Fig. 3	147	IN	10.0V	"		"	"	"	"	"	GND	"	GND	GND	"	A to J	9	180					"
"	"	148	10.0V	IN	"		"	"	"	"	"	"	"	GND	GND	"	B to J	"	"					"
"	"	149	GND	GND	"		"	"	"	"	"	"	"	IN	10.0V	"	C to J	"	"					"
"	"	150	"	"	"		"	"	"	"	"	"	"	10.0V	IN	"	D to J	"	"					"
"	"	151	"	"	"		IN	10.0V	"	"	"	"	"	GND	GND	"	E to J	"	"					"
"	"	152	"	"	"		10.0V	IN	"	"	"	"	"	"	"	"	F to J	"	"					"
"	"	153	"	"	"		GND	GND	"	IN	10.0V	"	"	"	"	"	G to J	"	"					"
"	"	154	"	"	"		"	"	"	10.0V	IN	"	"	"	"	"	H to J	"	"					"
"	"	155	"	"	"		"	"	"	GND	GND	"	IN	"	"	"	EN to J	"	"					"
t_{PLH}	"	156	"	"	"		"	"	"	"	"	IN	10.0V	"	"	"	I to J	10	200					"
"	"	157	IN	10.0V	"		"	"	"	"	"	GND	"	"	"	"	A to J	14	280					"
"	"	158	10.0V	IN	"		"	"	"	"	"	"	"	"	"	"	B to J	"	"					"
"	"	159	GND	GND	"		"	"	"	"	"	"	"	IN	10.0V	"	C to J	"	"					"
"	"	160	"	"	"		"	"	"	"	"	"	"	10.0V	IN	"	D to J	"	"					"
"	"	161	"	"	"		IN	10.0V	"	"	"	"	"	GND	GND	"	E to J	"	"					"
"	"	162	"	"	"		10.0V	IN	"	"	"	"	"	"	"	"	F to J	"	"					"
"	"	163	"	"	"		GND	GND	"	IN	10.0V	"	"	"	"	"	G to J	"	"					"
"	"	164	"	"	"		"	"	"	10.0V	IN	"	"	"	"	"	H to J	"	"					"
"	"	165	"	"	"		"	"	"	GND	GND	"	IN	"	"	"	EN to J	"	"					"
t_{THL}	3004	166	"	"	"		"	"	"	"	"	IN	10.0V	"	"	"	J	5	100					"
"	Fig. 3	167	IN	10.0V	"		"	"	"	"	"	GND	"	"	"	"	"	"	"					"
"	"	168	10.0V	IN	"		"	"	"	"	"	"	"	"	"	"	"	"	"					"
"	"	169	GND	GND	"		"	"	"	"	"	"	"	IN	10.0V	"	"	"	"					"
"	"	170	"	"	"		"	"	"	"	"	"	"	10.0V	IN	"	"	"	"					"
"	"	171	"	"	"		IN	10.0V	"	"	"	"	"	GND	GND	"	"	"	"					"
"	"	172	"	"	"		10.0V	IN	"	"	"	"	"	"	"	"	"	"	"					"
"	"	173	"	"	"		GND	GND	"	IN	10.0V	"	"	"	"	"	"	"	"					"
"	"	174	"	"	"		GND	GND	"	10.0V	IN	"	"	"	"	"	"	"	"					"
"	"	175	"	"	"		GND	GND	"	GND	GND	"	IN	"	"	"	"	"	"					"

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 – Continued.

Symbol	MIL-STD-833 method	Case A,C,D,T,X,Y	Terminal conditions 1/														Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 12 T_A = 25°C						
		Symbol	A	B	J	NC	E	F	V_{SS}	G	H	I	EN	C	D	V_{DD}								
		Test No.																Min	Max	Min	Max	Min	Max	
t_{TLH}	3004	176	GND	GND	OUT		GND	GND	GND	GND	GND	IN	10.0V	GND	GND	10.0V	J	5	100					ns
"	Fig. 3	177	IN	10.0V	"		"	"	"	"	"	GND	"	GND	GND	"	"	"	"					"
"	"	178	10.0V	IN	"		"	"	"	"	"	"	"	GND	GND	"	"	"	"					"
"	"	179	GND	GND	"		"	"	"	"	"	"	"	IN	10.0V	"	"	"	"					"
"	"	180	"	"	"		"	"	"	"	"	"	"	10.0V	IN	"	"	"	"					"
"	"	181	"	"	"		IN	10.0V	"	"	"	"	"	GND	GND	"	"	"	"					"
"	"	182	"	"	"		10.0V	IN	"	"	"	"	"	"	"	"	"	"	"					"
"	"	183	"	"	"		GND	GND	"	IN	10.0V	"	"	"	"	"	"	"	"					"
"	"	184	"	"	"		GND	GND	"	10.0V	IN	"	GND	"	"	"	"	"	"					"
"	"	185	"	"	"		GND	GND	"	GND	GND	"	IN	"	"	"	"	"	"					"

1/ Pins not designated may be "HIGH" level logic, "LOW" level logic or open. Exceptions are as follows: $V_{IC(pos)}$ tests, the V_{SS} terminals shall be open; $V_{IC(neg)}$ tests, the V_{DD} terminal shall be open; I_{SS} tests, the output terminals shall be open.
2/ The I_{SS} measurements shall be performed in sequence.
3/ The device manufacturer may, at his option, measure I_{IL} and I_{IH} at 25°C for each individual input or measure all inputs together.
4/ See 4.4.1c

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TABLE III. Group A inspection for device type 03.

Symbol	MIL-STD-833 method	Case A,C,D,T,X,Y	Terminal conditions 1/														Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	1A	1B	1Y	2Y	2A	2B	V_{SS}	3A	3B	3Y	4Y	4A	4B	V_{DD}								
		Test No.																Min	Max	Min	Max	Min	Max	
$V_{IC(pos)}$		1	1mA													GND	1A		1.5					Vdc
"		2		1mA												"	1B		"					"
"		3					1mA									"	2A		"					"
"		4						1mA								"	2B		"					"
"		5								1mA						"	3A		"					"
"		6									1mA					"	3B		"					"
"		7												1mA		"	4A		"					"
"		8													1mA	"	4B		"					"
$V_{IC(neg)}$		9	-1mA						GND							GND	1A		-6.0					"
"		10		-1mA					"							"	1B		"					"
"		11					-1mA		"							"	2A		"					"
"		12						-1mA	"							"	2B		"					"
"		13							"	-1mA						"	3A		"					"
"		14							"		-1mA					"	3B		"					"
"		15							"					-1mA		"	4A		"					"
"		16							"						-1mA	"	4B		"					"
I_{SS}	3005	17	GND	GND			GND	GND	"	GND	GND			GND	GND	18.0V	V_{SS}		-25		-750			nA
2/	"	18	GND	18.0V			GND	18.0V	"	GND	18.0V			GND	18.0V	"	"		"		"			"
"	"	19	18.0V	GND			18.0V	GND	"	18.0V	GND			18.0V	GND	"	"		"		"			"
"	"	20	18.0V	18.0V			18.0V	18.0V	"	18.0V	18.0V			18.0V	18.0V	"	"		"		"			"
V_{OL1}	3007	21	GND	GND			GND	GND	"	GND	GND			GND	GND	15.0V	1Y		0.05		0.05		0.05	Vdc
"	"	22	"	"			"	"	"	"	"			"	"	"	2Y		"		"		"	"
"	"	23	"	"			"	"	"	"	"			"	"	"	3Y		"		"		"	"
"	"	24	"	"			"	"	"	"	"			"	"	"	4Y		"		"		"	"
"	"	25	15.0V	15.0V			15.0V	15.0V	"	15.0V	15.0V			15.0V	15.0V	"	1Y		"		"		"	"
"	"	26	"	"			"	"	"	"	"			"	"	"	2Y		"		"		"	"
"	"	27	"	"			"	"	"	"	"			"	"	"	3Y		"		"		"	"
"	"	28	"	"			"	"	"	"	"			"	"	"	4Y		"		"		"	"
V_{OH1}	3006	29	"	GND			GND	GND	"	GND	GND			GND	GND	"	1Y	14.95		14.95		14.95		"
"	"	30	GND	15.0V			GND	GND	"	"	"			"	"	"	1Y	"		"		"		"
"	"	31	"	GND			15.0V	GND	"	"	"			"	"	"	2Y	"		"		"		"
"	"	32	"	"			GND	15.0V	"	"	"			"	"	"	2Y	"		"		"		"
"	"	33	"	"			"	GND	"	15.0V	"			"	"	"	3Y	"		"		"		"
"	"	34	"	"			"	"	"	GND	15.0V			"	"	"	3Y	"		"		"		"
"	"	35	"	"			"	"	"	"	GND			15.0V	"	"	4Y	"		"		"		"
"	"	36	"	"			"	"	"	"	"			GND	15.0V	"	4Y	"		"		"		"
V_{IH1}		37	3.5V	1.5V			"	"	"	"	"			"	GND	5.0V	1Y	4.5		4.5		4.5		"
"		38	1.5V	3.5V			"	"	"	"	"			"	"	"	1Y	"		"		"		"
"		39	GND	GND			3.5V	1.5V	"	"	"			"	"	"	2Y	"		"		"		"
"		40	"	"			1.5V	3.5V	"	"	"			"	"	"	2Y	"		"		"		"
"		41	"	"			GND	GND	"	3.5V	1.5V			"	"	"	3Y	"		"		"		"
"		42	"	"			"	"	"	1.5V	3.5V			"	"	"	3Y	"		"		"		"
"		43	"	"			"	"	"	GND	GND			3.5V	1.5V	"	4Y	"		"		"		"
"		44	"	"			"	"	"	GND	GND			1.5V	3.5V	"	4Y	"		"		"		"

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03 – Continued.

Symbol	MIL-STD-833 method	Case A,C,D,T,X,Y	Terminal conditions 1/														Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	1A	1B	1Y	2Y	2A	2B	V_{SS}	3A	3B	3Y	4Y	4A	4B	V_{DD}								
		Test No.																Min	Max	Min	Max	Min	Max	
V_{IH2}		45	7.0V	3.0V			GND	GND	GND	GND	GND			GND	GND	10.0V	1Y	9.0V		9.0V		9.0V		Vdc
"		46	3.0V	7.0V			GND	GND	"	"	"			"	"	"	1Y	"		"		"		"
"		47	GND	GND			7.0V	3.0V	"	"	"			"	"	"	2Y	"		"		"		"
"		48	"	"			3.0V	7.0V	"	"	"			"	"	"	2Y	"		"		"		"
"		49	"	"			GND	GND	"	7.0V	3.0V			"	"	"	3Y	"		"		"		"
"		50	"	"			"	"	"	3.0V	7.0V			"	"	"	3Y	"		"		"		"
"		51	"	"			"	"	"	GND	GND			7.0V	3.0V	"	4Y	"		"		"		"
"		52	"	"			"	"	"	"	"			3.0V	7.0V	"	4Y	"		"		"		"
V_{IH3}		53	11.0V	4.0V			"	"	"	"	"			GND	GND	15.0V	1Y	13.5		13.5		13.5		"
"		54	4.0V	11.0V			"	"	"	"	"			"	"	"	1Y	"		"		"		"
"		55	GND	GND			11.0V	4.0V	"	"	"			"	"	"	2Y	"		"		"		"
"		56	"	"			4.0V	11.0V	"	"	"			"	"	"	2Y	"		"		"		"
"		57	"	"			GND	GND	"	11.0V	4.0V			"	"	"	3Y	"		"		"		"
"		58	"	"			"	"	"	4.0V	11.0V			"	"	"	3Y	"		"		"		"
"		59	"	"			"	"	"	GND	GND			11.0V	4.0V	"	4Y	"		"		"		"
"		60	"	"			"	"	"	"	"			4.0V	11.0V	"	4Y	"		"		"		"
V_{IL1}		61	1.5V	"			"	"	"	"	"			GND	GND	5.0V	1Y		0.5		0.5		0.5	"
"		62	GND	"			1.5V	"	"	"	"			GND	"	"	2Y		"		"		"	"
"		63	GND	"			GND	"	"	1.5V	"			GND	"	"	3Y		"		"		"	"
"		64	GND	"			GND	"	"	GND	"			1.5V	"	"	4Y		"		"		"	"
V_{IL2}		65	3.0V	"			GND	"	"	GND	"			GND	"	10.0V	1Y		1.0		1.0		1.0	"
"		66	GND	"			3.0V	"	"	GND	"			GND	"	"	2Y		"		"		"	"
"		67	GND	"			GND	"	"	3.0V	"			GND	"	"	3Y		"		"		"	"
"		68	GND	"			GND	"	"	GND	"			3.0V	"	"	4Y		"		"		"	"
V_{IL3}		69	4.0V	"			GND	"	"	GND	"			GND	"	15.0V	1Y		1.5		1.5		1.5	"
"		70	GND	"			4.0V	"	"	GND	"			GND	"	"	2Y		"		"		"	"
"		71	GND	"			GND	"	"	4.0V	"			GND	"	"	3Y		"		"		"	"
"		72	GND	"			GND	"	"	GND	"			4.0V	"	"	4Y		"		"		"	"
I_{IH1} 3/	3010	73	18.0V	18.0V			18.0V	18.0V	"	18.0V	18.0V			18.0V	18.0V	18.0V	All inputs together		800					nA
I_{IH2}	"	74	18.0V	GND			GND	GND	"	GND	GND			GND	GND	"	1A		100		100			"
"	"	75	GND	18.0V			GND	GND	"	"	"			"	"	"	1B		"		"			"
"	"	76	"	GND			18.0V	GND	"	"	"			"	"	"	2A		"		"			"
"	"	77	"	"			GND	18.0V	"	"	"			"	"	"	2B		"		"			"
"	"	78	"	"			"	GND	"	18.0V	"			"	"	"	3A		"		"			"
"	"	79	"	"			"	"	"	GND	18.0V			"	"	"	3B		"		"			"
"	"	80	"	"			"	"	"	"	GND			18.0V	"	"	4A		"		"			"
"	"	81	"	"			"	"	"	"	GND			GND	18.0V	"	4B		"		"			"
I_{IL1} 3/	3009	82	"	"			"	"	"	GND	GND			GND	GND	"	All inputs together		-800					"
I_{IL2}	"	83	"	18.0V			18.0V	18.0V	"	18.0V	18.0V			18.0V	18.0V	"	1A		-100		-100			"
"	"	84	18.0V	GND			18.0V	18.0V	"	"	"			"	"	"	1B		"		"			"
"	"	85	"	18.0V			GND	18.0V	"	"	"			"	"	"	2A		"		"			"
"	"	86	"	"			18.0V	GND	"	"	"			"	"	"	2B		"		"			"
"	"	87	"	"			"	18.0V	"	GND	"			"	"	"	3A		"		"			"
"	"	88	"	"			"	"	"	18.0V	GND			"	"	"	3B		"		"			"
"	"	89	"	"			"	"	"	18.0V	18.0V			GND	"	"	4A		"		"			"
"	"	90	"	"			"	"	"	18.0V	18.0V			18.0V	GND	"	4B		"		"			"

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03 – Continued.

Symbol	MIL-STD-833 method	Case A,C,D,T,X,Y Symbol	Terminal conditions 1/														Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
			1A	1B	1Y	2Y	2A	2B	V_{SS}	3A	3B	3Y	4Y	4A	4B	V_{DD}								
		Test No.																Min	Max	Min	Max	Min	Max	
I_{OL1}		91	GND	GND	0.4V		GND	GND	GND	GND	GND			GND	GND	5.0V	1Y	0.51		0.36		0.64		mA
"		92	"	"		0.4V	"	"	"	"	"			"	"	"	2Y	"		"		"		"
"		93	"	"			"	"	"	"	"	0.4V		"	"	"	3Y	"		"		"		"
"		94	"	"			"	"	"	"	"		0.4V	"	"	"	4Y	"		"		"		"
I_{OH1}		95	5.0V	"	4.6V		5.0V	"	"	5.0V	"			5.0V	"	"	1Y	-0.51		-0.36		-0.64		"
"		96	"	"		4.6V	"	"	"	"	"			"	"	"	2Y	"		"		"		"
"		97	"	"			"	"	"	"	"	4.6V		"	"	"	3Y	"		"		"		"
"		98	"	"			"	"	"	"	"		4.6V	"	"	"	4Y	"		"		"		"
I_{OL2}		99	GND	"	1.5V		GND	"	"	GND	"			GND	"	15.0V	1Y	3.4		2.4		4.2		"
"		100	"	"		1.5V	"	"	"	"	"			"	"	"	2Y	"		"		"		"
"		101	"	"			"	"	"	"	"	1.5V		"	"	"	3Y	"		"		"		"
"		102	"	"			"	"	"	"	"		1.5V	"	"	"	4Y	"		"		"		"
I_{OH2}		103	15.0V	"	13.5V		15.0V	"	"	15.0V	"			15.0V	"	"	1Y	-3.4		-2.4		-4.2		"
"		104	"	"		13.5V	"	"	"	"	"			"	"	"	2Y	"		"		"		"
"		105	"	"			"	"	"	"	"	13.5V		"	"	"	3Y	"		"		"		"
"		106	"	"			"	"	"	"	"		13.5V	"	"	"	4Y	"		"		"		"
																		Subgroup 4 T_A = 25°C						
																		Min	Max					
C_I	3012	107	4/						GND							GND	1A		7.5					pF
"	"	108		4/					"							"	1B		"					"
"	"	109					4/		"							"	2A		"					"
"	"	110						4/	"							"	2B		"					"
"	"	111							"	4/						"	3A		"					"
"	"	112							"		4/					"	3B		"					"
"	"	113							"					4/		"	4A		"					"
"	"	114							"						4/	"	4B		"					"
																		Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
																		Min	Max	Min	Max	Min	Max	
t_{PHL}	3003	115	IN	GND	OUT		GND	GND	GND	GND	GND			GND	GND	5.0V	1A to 1Y	13	250	18	350	13	250	ns
"	Fig. 3	116	GND	IN	OUT		GND	GND	"	"	"			"	"	"	1B to 1Y	"	"	"	"	"	"	"
"	"	117	"	GND		OUT	IN	GND	"	"	"			"	"	"	2A to 2Y	"	"	"	"	"	"	"
"	"	118	"	"		OUT	GND	IN	"	"	"			"	"	"	2B to 2Y	"	"	"	"	"	"	"
"	"	119	"	"			"	GND	"	IN	"	OUT		"	"	"	3A to 3Y	"	"	"	"	"	"	"
"	"	120	"	"			"	"	"	GND	IN	OUT		"	"	"	3B to 3Y	"	"	"	"	"	"	"
"	"	121	"	"			"	"	"	"	GND		OUT	IN	"	"	4A to 4Y	"	"	"	"	"	"	"
"	"	122	"	"			"	"	"	"	"		OUT	GND	IN	"	4B to 4Y	"	"	"	"	"	"	"
t_{PLH}	3006	123	IN	"	OUT		"	"	"	"	"			"	GND	"	1A to 1Y	"	"	"	"	"	"	"
"	Fig. 3	124	GND	IN	OUT		"	"	"	"	"			"	"	"	1B to 1Y	"	"	"	"	"	"	"
"	"	125	"	GND		OUT	IN	"	"	"	"			"	"	"	2A to 2Y	"	"	"	"	"	"	"
"	"	126	"	"		OUT	GND	IN	"	"	"			"	"	"	2B to 2Y	"	"	"	"	"	"	"
"	"	127	"	"			"	GND	"	IN	"	OUT		"	"	"	3A to 3Y	"	"	"	"	"	"	"
"	"	128	"	"			"	"	"	GND	IN	OUT		"	"	"	3B to 3Y	"	"	"	"	"	"	"
"	"	129	"	"			"	"	"	GND	GND		OUT	IN	"	"	4A to 4Y	"	"	"	"	"	"	"
"	"	130	"	"			"	"	"	GND	GND		OUT	GND	IN	"	4B to 4Y	"	"	"	"	"	"	"

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03 – Continued.

Symbol	MIL-STD-833 method	Case A,C,D,T,X,Y Symbol	Terminal conditions 1/														Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
			1A	1B	1Y	2Y	2A	2B	V_{SS}	3A	3B	3Y	4Y	4A	4B	V_{DD}								
		Test No.																Min	Max	Min	Max	Min	Max	
t_{THL}	3004	131	IN	GND	OUT		GND	GND	GND	GND	GND			GND	GND	5.0V	1Y	10	200	18	280	10	200	ns
"	Fig. 3	132	GND	IN	OUT		GND	GND	"	"	"			"	"	"	1Y	"	"	"	"	"	"	"
"	"	133	"	GND		OUT	IN	GND	"	"	"			"	"	"	2Y	"	"	"	"	"	"	"
"	"	134	"	"		OUT	GND	IN	"	"	"			"	"	"	2Y	"	"	"	"	"	"	"
"	"	135	"	"			"	GND	"	IN	"	OUT		"	"	"	3Y	"	"	"	"	"	"	"
"	"	136	"	"			"	"	"	GND	IN	OUT		"	"	"	3Y	"	"	"	"	"	"	"
"	"	137	"	"			"	"	"	"	GND		OUT	IN	"	"	4Y	"	"	"	"	"	"	"
"	"	138	"	"			"	"	"	"	"		OUT	GND	IN	"	4Y	"	"	"	"	"	"	"
t_{TLH}	"	139	IN	"	OUT		"	"	"	"	"			GND	GND	"	1Y	"	"	"	"	"	"	"
"	"	140	GND	IN	OUT		"	"	"	"	"			"	"	"	1Y	"	"	"	"	"	"	"
"	"	141	"	GND		OUT	IN	"	"	"	"			"	"	"	2Y	"	"	"	"	"	"	"
"	"	142	"	"		OUT	GND	IN	"	"	"			"	"	"	2Y	"	"	"	"	"	"	"
"	"	143	"	"			"	GND	"	IN	"	OUT		"	"	"	3Y	"	"	"	"	"	"	"
"	"	144	"	"			"	"	"	GND	IN	OUT		"	"	"	3Y	"	"	"	"	"	"	"
"	"	145	"	"			"	"	"	GND	GND		OUT	IN	"	"	4Y	"	"	"	"	"	"	"
"	"	146	"	"			"	"	"	GND	GND		OUT	GND	IN	"	4Y	"	"	"	"	"	"	"
																		Subgroup 12 T_A = 25°C						
																		Min	Max					
t_{PHL}	3003	147	IN	GND	OUT		GND	GND	GND	GND	GND			GND	GND	10.0V	1A to 1Y	6	120					ns
"	Fig. 3	148	GND	IN	OUT		GND	GND	"	"	"			"	"	"	1B to 1Y	"	"					"
"	"	149	"	GND		OUT	IN	GND	"	"	"			"	"	"	2A to 2Y	"	"					"
"	"	150	"	"		OUT	GND	IN	"	"	"			"	"	"	2B to 2Y	"	"					"
"	"	151	"	"			"	GND	"	IN	"	OUT		"	"	"	3A to 3Y	"	"					"
"	"	152	"	"			"	"	"	GND	IN	OUT		"	"	"	3B to 3Y	"	"					"
"	"	153	"	"			"	"	"	"	GND		OUT	IN	"	"	4A to 4Y	"	"					"
"	"	154	"	"			"	"	"	"	"		OUT	GND	IN	"	4B to 4Y	"	"					"
t_{PLH}	"	155	IN	"	OUT		"	"	"	"	"			"	GND	10.0V	1A to 1Y	"	"					"
"	"	156	GND	IN	OUT		"	"	"	"	"			"	"	"	1B to 1Y	"	"					"
"	"	157	"	GND		OUT	IN	"	"	"	"			"	"	"	2A to 2Y	"	"					"
"	"	158	"	"		OUT	GND	IN	"	"	"			"	"	"	2B to 2Y	"	"					"
"	"	159	"	"			"	GND	"	IN	"	OUT		"	"	"	3A to 3Y	"	"					"
"	"	160	"	"			"	"	"	GND	IN	OUT		"	"	"	3B to 3Y	"	"					"
"	"	161	"	"			"	"	"	"	GND		OUT	IN	"	"	4A to 4Y	"	"					"
"	"	162	"	"			"	"	"	"	"		OUT	GND	IN	"	4B to 4Y	"	"					"
t_{THL}	3004	163	IN	"	OUT		"	"	"	"	"			GND	GND	"	1Y	5	100					"
"	Fig. 3	164	GND	IN	OUT		"	"	"	"	"			"	"	"	1Y	"	"					"
"	"	165	"	GND		OUT	IN	"	"	"	"			"	"	"	2Y	"	"					"
"	"	166	"	"		OUT	GND	IN	"	"	"			"	"	"	2Y	"	"					"
"	"	167	"	"			"	GND	"	IN	"	OUT		"	"	"	3Y	"	"					"
"	"	168	"	"			"	"	"	GND	IN	OUT		"	"	"	3Y	"	"					"
"	"	169	"	"			"	"	"	GND	GND		OUT	IN	"	"	4Y	"	"					"
"	"	170	"	"			"	"	"	GND	GND		OUT	GND	IN	"	4Y	"	"					"

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03 – Continued.

Symbol	MIL-STD-833 method	Case A,C,D,T,X,Y Symbol	Terminal conditions 1/														Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 12 T_A = 25°C						
			1A	1B	1Y	2Y	2A	2B	V_{SS}	3A	3B	3Y	4Y	4A	4B	V_{DD}								
		Test No.																Min	Max	Min	Max	Min	Max	
t_{TLH}	3004	171	IN	GND	OUT		GND	GND	GND	GND	GND			GND	GND	10.0V	1Y	5	100					ns
"	Fig. 3	172	GND	IN	OUT		GND	GND	"	"	"			"	"	"	1Y	"	"					"
"	"	173	"	GND		OUT	IN	GND	"	"	"			"	"	"	2Y	"	"					"
"	"	174	"	"		OUT	GND	IN	"	"	"			"	"	"	2Y	"	"					"
"	"	175	"	"			"	GND	"	IN	"	OUT		"	"	"	3Y	"	"					"
"	"	176	"	"			"	"	"	GND	IN	OUT		"	"	"	3Y	"	"					"
"	"	177	"	"			"	"	"	GND	GND		OUT	IN	"	"	4Y	"	"					"
"	"	178	"	"			"	"	"	GND	GND		OUT	GND	IN	"	4Y	"	"					"

1/ Pins not designated may be "HIGH" level logic, "LOW" level logic or open. Exceptions are as follows: $V_{IC(pos)}$ tests, the V_{SS} terminals shall be open: $V_{IC(neg)}$ tests, the V_{DD} terminal shall be open; I_{SS} tests, the output terminals shall be open.
2/ The I_{SS} measurements shall be performed in sequence.
3/ The device manufacturer may, at his option, measure I_{IL} and I_{IH} at 25°C for each individual input or measure all inputs together.
4/ See 4.4.1c.

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TABLE III. Group A inspection for device type 04.

Symbol	MIL-STD-833 method	Case A,C,D,T,X,Y	Terminal conditions 1/														Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	1A	1B	1Y	2Y	2A	2B	V_{SS}	4A	4B	4Y	3Y	3A	3B	V_{DD}								
		Test No.																Min	Max	Min	Max	Min	Max	
V_{IC} (pos)		1	1mA													GND	1A		1.5					Vdc
"		2		1mA												"	1B		"					"
"		3					1mA									"	2A		"					"
"		4						1mA								"	2B		"					"
"		5								1mA						"	4A		"					"
"		6									1mA					"	4B		"					"
"		7												1mA		"	3A		"					"
"		8													1mA	"	3B		"					"
V_{IC} (neg)		9	-1mA						GND								1A		-6.0					"
"		10		-1mA					"								1B		"					"
"		11					-1mA		"								2A		"					"
"		12						-1mA	"								2B		"					"
"		13							"	-1mA							4A		"					"
"		14							"		-1mA						4B		"					"
"		15							"					-1mA			3A		"					"
"		16							"						-1mA		3B		"					"
I_{SS} 2/	3005	17	GND	GND			GND	GND	"	GND	GND			GND	GND	18.0V	V_{SS}		-25		-750			nA
"	"	18	GND	18.0V			GND	18.0V	"	GND	18.0V			GND	18.0V	"	"		"		"			"
"	"	19	18.0V	GND			18.0V	GND	"	18.0V	GND			18.0V	GND	"	"		"		"			"
"	"	20	18.0V	18.0V			18.0V	18.0V	"	18.0V	18.0V			18.0V	18.0V	"	"		"		"			"
V_{OH1}	3006	21	GND	GND			GND	GND	"	GND	GND			GND	GND	15.0V	1Y	14.95		14.95		14.95		Vdc
"	"	22	"	"			"	"	"	"	"			"	"	"	2Y	"		"		"		"
"	"	23	"	"			"	"	"	"	"			"	"	"	4Y	"		"		"		"
"	"	24	"	"			"	"	"	"	"			"	"	"	3Y	"		"		"		"
"	"	25	15.0V	15.0V			15.0V	15.0V	"	15.0V	15.0V			15.0V	15.0V	"	1Y	"		"		"		"
"	"	26	"	"			"	"	"	"	"			"	"	"	2Y	"		"		"		"
"	"	27	"	"			"	"	"	"	"			"	"	"	4Y	"		"		"		"
"	"	28	"	"			"	"	"	"	"			"	"	"	3Y	"		"		"		"
V_{OL1}	3007	29	"	GND			GND	GND	"	GND	GND			GND	GND	"	1Y		0.05		0.05		0.05	"
"	"	30	GND	15.0V			GND	GND	"	"	"			"	"	"	2Y		"		"		"	"
"	"	31	"	GND			15.0V	GND	"	"	"			"	"	"	4Y		"		"		"	"
"	"	32	"	"			GND	15.0V	"	"	"			"	"	"	3Y		"		"		"	"
"	"	33	"	"			"	GND	"	15.0V	"			"	"	"	1Y		"		"		"	"
"	"	34	"	"			"	"	"	GND	15.0V			"	"	"	2Y		"		"		"	"
"	"	35	"	"			"	"	"	"	GND			15.0V	"	"	4Y		"		"		"	"
"	"	36	"	"			"	"	"	"	"			GND	15.0V	"	3Y		"		"		"	"
V_{IH1}		37	3.5V	1.5V			"	"	"	"	"			"	GND	5.0V	1Y		0.05		0.05		0.05	"
"		38	1.5V	3.5V			"	"	"	"	"			"	"	"	1Y		"		"		"	"
"		39	GND	GND			3.5V	1.5V	"	"	"			"	"	"	2Y		"		"		"	"
"		40	"	"			1.5V	3.5V	"	"	"			"	"	"	2Y		"		"		"	"
"		41	"	"			GND	GND	"	3.5V	1.5V			"	"	"	4Y		"		"		"	"
"		42	"	"			"	"	"	1.5V	3.5V			"	"	"	4Y		"		"		"	"
"		43	"	"			"	"	"	GND	GND			3.5V	1.5V	"	3Y		"		"		"	"
"		44	"	"			"	"	"	GND	GND			1.5V	3.5V	"	3Y		"		"		"	"

See footnotes at end of device type 04.

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TABLE III. Group A inspection for device type 04 – Continued.

Symbol	MIL-STD-833 method	Case A,C,D,T,X,Y	Terminal conditions 1/														Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	1A	1B	1Y	2Y	2A	2B	V_{SS}	4A	4B	4Y	3Y	3A	3B	V_{DD}								
		Test No.																Min	Max	Min	Max	Min	Max	
V_{IH2}		45	7.0V	3.0V			GND	GND	GND	GND	GND			GND	GND	10.0V	1Y		1.0		1.0		1.0	Vdc
"		46	3.0V	7.0V			GND	GND	"	"	"			"	"	"	1Y		"		"		"	"
"		47	GND	GND			7.0V	3.0V	"	"	"			"	"	"	2Y		"		"		"	"
"		48	"	"			3.0V	7.0V	"	"	"			"	"	"	2Y		"		"		"	"
"		49	"	"			GND	GND	"	7.0V	3.0V			"	"	"	4Y		"		"		"	"
"		50	"	"			"	"	"	3.0V	7.0V			"	"	"	4Y		"		"		"	"
"		51	"	"			"	"	"	GND	GND			7.0V	3.0V	"	3Y		"		"		"	"
"		52	"	"			"	"	"	"	"			3.0V	7.0V	"	3Y		"		"		"	"
V_{IH3}		53	11.0V	4.0V			"	"	"	"	"			GND	GND	15.0V	1Y		1.5		1.5		1.5	"
"		54	4.0V	11.0V			"	"	"	"	"			"	"	"	1Y		"		"		"	"
"		55	GND	GND			11.0V	4.0V	"	"	"			"	"	"	2Y		"		"		"	"
"		56	"	"			4.0V	11.0V	"	"	"			"	"	"	2Y		"		"		"	"
"		57	"	"			GND	GND	"	11.0V	4.0V			"	"	"	4Y		"		"		"	"
"		58	"	"			"	"	"	4.0V	11.0V			"	"	"	4Y		"		"		"	"
"		59	"	"			"	"	"	GND	GND			11.0V	4.0V	"	3Y		"		"		"	"
"		60	"	"			"	"	"	"	"			4.0V	11.0V	"	3Y		"		"		"	"
V_{IL1}		61	1.5V	"			"	"	"	"	"			GND	GND	5.0V	1Y	4.5		4.5		4.5		"
"		62	GND	"			1.5V	"	"	"	"			GND	"	"	2Y	"		"		"		"
"		63	GND	"			GND	"	"	1.5V	"			GND	"	"	4Y	"		"		"		"
"		64	GND	"			GND	"	"	GND	"			1.5V	"	"	3Y	"		"		"		"
V_{IL2}		65	3.0V	"			GND	"	"	GND	"			GND	GND	10.0V	1Y	9.0		9.0		9.0		"
"		66	GND	"			3.0V	"	"	GND	"			GND	"	"	2Y	"		"		"		"
"		67	GND	"			GND	"	"	3.0V	"			GND	"	"	4Y	"		"		"		"
"		68	GND	"			GND	"	"	GND	"			3.0V	"	"	3Y	"		"		"		"
V_{IL3}		69	4.0V	"			GND	"	"	GND	"			GND	GND	15.0V	1Y	13.5		13.5		13.5		"
"		70	GND	"			4.0V	"	"	GND	"			GND	"	"	2Y	"		"		"		"
"		71	GND	"			GND	"	"	4.0V	"			GND	"	"	4Y	"		"		"		"
"		72	GND	"			GND	"	"	GND	"			4.0V	"	"	3Y	"		"		"		"
I_{IH1} 3/	3010	73	18.0V	18.0V			18.0V	18.0V	"	18.0V	18.0V			18.0V	18.0V	18.0V	All inputs together		8					nA
I_{IH2}	"	74	18.0V	GND			GND	GND	"	GND	GND			GND	GND	"	1A		1		45			"
"	"	75	GND	18.0V			GND	GND	"	"	"			"	"	"	1B		"		"			"
"	"	76	"	GND			18.0V	GND	"	"	"			"	"	"	2A		"		"			"
"	"	77	"	"			GND	18.0V	"	"	"			"	"	"	2B		"		"			"
"	"	78	"	"			"	GND	"	18.0V	"			"	"	"	4A		"		"			"
"	"	79	"	"			"	"	"	GND	18.0V			"	"	"	4B		"		"			"
"	"	80	"	"			"	"	"	"	GND			18.0V	"	"	3A		"		"			"
"	"	81	"	"			"	"	"	"	GND			GND	18.0V	"	3B		"		"			"
I_{IL1} 3/	3009	82	"	"			"	"	"	"	GND			GND	GND	GND	All inputs together		-8					"

See footnotes at end of device type 04.

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TABLE III. Group A inspection for device type 04 – Continued.

Symbol	MIL-STD-833 method	Case A,C,D,T,X,Y	Terminal conditions 1/														Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Symbol	1A	1B	1Y	2Y	2A	2B	V_{SS}	4A	4B	4Y	3Y	3A	3B	V_{DD}								
		Test No.																Min	Max	Min	Max	Min	Max	
I_{IL2}	3009	83	GND	18.0V			18.0V	18.0V	GND	18.0V	18.0V			18.0V	18.0V	18.0V	1A		-1		-45		"	nA
"	"	84	18.0V	GND			18.0V	18.0V	"	"	"			"	"	"	1B		"		"		"	"
"	"	85	"	18.0V			GND	18.0V	"	"	"			"	"	"	2A		"		"		"	"
"	"	86	"	"			18.0V	GND	"	"	"			"	"	"	2B		"		"		"	"
"	"	87	"	"			"	18.0V	"	GND	"			"	"	"	4A		"		"		"	"
"	"	88	"	"			"	18.0V	"	18.0V	GND			"	"	"	4B		"		"		"	"
"	"	89	"	"			"	18.0V	"	18.0V	18.0V			GND	"	"	3A		"		"		"	"
"	"	90	"	"			"	GND	"	18.0V	18.0V			18.0V	GND	"	3B		"		"		"	"
I_{OH1}		91	GND	GND	4.6V		GND	"	"	GND	GND			GND	"	5.0V	1Y	-0.51		-0.36		-0.64		mA
"		92	"	"		4.6V	"	"	"	"	"			"	"	"	2Y	"		"		"		"
"		93	"	"			"	"	"	"	"	4.6V		"	"	"	4Y	"		"		"		"
"		94	"	"			"	"	"	"	"		4.6V	"	"	"	3Y	"		"		"		"
I_{OL1}		95	5.0V	"	0.4V		5.0V	"	"	5.0V	"			5.0V	"	"	1Y	0.51		0.36		0.64		"
"		96	"	"		0.4V	"	"	"	"	"			"	"	"	2Y	"		"		"		"
"		97	"	"			"	"	"	"	"	0.4V		"	"	"	4Y	"		"		"		"
"		98	"	"			"	"	"	"	"		0.4V	"	"	"	3Y	"		"		"		"
I_{OH2}		99	GND	"	13.5V		GND	"	"	GND	"			GND	"	15.0V	1Y	-3.4		-2.4		-4.2		"
"		100	"	"		13.5V	"	"	"	"	"			"	"	"	2Y	"		"		"		"
"		101	"	"			"	"	"	"	"	13.5V		"	"	"	4Y	"		"		"		"
"		102	"	"			"	"	"	"	"		13.5V	"	"	"	3Y	"		"		"		"
I_{OL2}		103	15.0V	"	1.5V		15.0V	"	"	15.0V	"			15.0V	"	"	1Y	3.4		2.4		4.2		"
"		104	"	"		1.5V	"	"	"	"	"			"	"	"	2Y	"		"		"		"
"		105	"	"			"	"	"	"	"	1.5V		"	"	"	4Y	"		"		"		"
"		106	"	"			"	"	"	"	"		1.5V	"	"	"	3Y	"		"		"		"
																		Subgroup 4 T_A = 25°C						
																		Min	Max					
C_i	3012	107	4/						GND							GND	1A		7.5					pF
"	"	108		4/					"							"	1B		"					"
"	"	109					4/		"							"	2A		"					"
"	"	110						4/	"							"	2B		"					"
"	"	111							"	4/						"	4A		"					"
"	"	112							"		4/					"	4B		"					"
"	"	113							"					4/		"	3A		"					"
"	"	114							"						4/	"	3B		"					"
																		Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
																		Min	Max	Min	Max	Min	Max	
t_{PHL}	3003	115	IN	GND	OUT		GND	GND	GND	GND	GND			GND	GND	5.0V	1A to 1Y	13	250	18	350	13	250	ns
"	Fig. 3	116	GND	IN	OUT		GND	GND	"	"	"			"	"	"	1B to 1Y	"	"	"	"	"	"	"
"	"	117	"	GND		OUT	IN	GND	"	"	"			"	"	"	2A to 2Y	"	"	"	"	"	"	"
"	"	118	"	"		OUT	GND	IN	"	"	"			"	"	"	2B to 2Y	"	"	"	"	"	"	"
"	"	119	"	"			"	GND	"	IN	"	OUT		"	"	"	4A to 4Y	"	"	"	"	"	"	"
"	"	120	"	"			"	"	"	GND	IN	OUT		"	"	"	4B to 4Y	"	"	"	"	"	"	"
"	"	121	"	"			"	"	"	GND	GND		OUT	IN	"	"	3A to 3Y	"	"	"	"	"	"	"
"	"	122	"	"			"	"	"	GND	GND		OUT	GND	IN	"	3B to 3Y	"	"	"	"	"	"	"

See footnotes at end of device type 04.

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TABLE III. Group A inspection for device type 04 – Continued.

Symbol	MIL-STD-833 method	Case A,C,D,T,X,Y	Terminal conditions 1/														Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
		Symbol	1A	1B	1Y	2Y	2A	2B	V_{SS}	4A	4B	4Y	3Y	3A	3B	V_{DD}								
		Test No.																Min	Max	Min	Max	Min	Max	
t_{PLH}	3003	123	IN	GND	OUT		GND	GND	GND	GND	GND			GND	GND	5.0V	1A to 1Y	13	250	18	350	13	250	ns
"	Fig. 3	124	GND	IN	OUT		GND	GND	"	"	"			"	"	"	1B to 1Y	"	"	"	"	"	"	"
"	"	125	"	GND		OUT	IN	GND	"	"	"			"	"	"	2A to 2Y	"	"	"	"	"	"	"
"	"	126	"	"		OUT	GND	IN	"	"	"			"	"	"	2B to 2Y	"	"	"	"	"	"	"
"	"	127	"	"			"	GND	"	IN	"	OUT		"	"	"	4A to 4Y	"	"	"	"	"	"	"
"	"	128	"	"			"	"	"	GND	IN	OUT		"	"	"	4B to 4Y	"	"	"	"	"	"	"
"	"	129	"	"			"	"	"	"	GND		OUT	IN	"	"	3A to 3Y	"	"	"	"	"	"	"
"	"	130	"	"			"	"	"	"	"		OUT	GND	IN	"	3B to 3Y	"	"	"	"	"	"	"
t_{THL}	3004	131	IN	"	OUT		"	"	"	"	"			"	GND	"	1Y	10	200	14	280	10	200	"
"	Fig. 3	132	GND	IN	OUT		"	"	"	"	"			"	"	"	1Y	"	"	"	"	"	"	"
"	"	133	"	GND		OUT	IN	"	"	"	"			"	"	"	2Y	"	"	"	"	"	"	"
"	"	134	"	"		OUT	GND	IN	"	"	"			"	"	"	2Y	"	"	"	"	"	"	"
"	"	135	"	"			"	GND	"	IN	"	OUT		"	"	"	4Y	"	"	"	"	"	"	"
"	"	136	"	"			"	"	"	GND	IN	OUT		"	"	"	4Y	"	"	"	"	"	"	"
"	"	137	"	"			"	"	"	"	GND		OUT	IN	"	"	3Y	"	"	"	"	"	"	"
"	"	138	"	"			"	"	"	"	"		OUT	GND	IN	"	3Y	"	"	"	"	"	"	"
t_{TLH}	"	139	IN	"	OUT		"	"	"	"	"			"	GND	"	1Y	"	"	"	"	"	"	"
"	"	140	GND	IN	OUT		"	"	"	"	"			"	"	"	1Y	"	"	"	"	"	"	"
"	"	141	"	GND		OUT	IN	"	"	"	"			"	"	"	2Y	"	"	"	"	"	"	"
"	"	142	"	"		OUT	GND	IN	"	"	"			"	"	"	2Y	"	"	"	"	"	"	"
"	"	143	"	"			"	GND	"	IN	"	OUT		"	"	"	4Y	"	"	"	"	"	"	"
"	"	144	"	"			"	"	"	GND	IN	OUT		"	"	"	4Y	"	"	"	"	"	"	"
"	"	145	"	"			"	"	"	GND	GND		OUT	IN	"	"	3Y	"	"	"	"	"	"	"
"	"	146	"	"			"	"	"	GND	GND		OUT	GND	IN	"	3Y	"	"	"	"	"	"	"
																		Subgroup 12 T_A = 25°C						
																		Min	Max					
t_{PHL}	3003	147	IN	GND	OUT		GND	GND	GND	GND	GND			GND	GND	10.0V	1A to 1Y	6	120					ns
"	Fig. 3	148	GND	IN	OUT		GND	GND	"	"	"			"	"	"	1B to 1Y	"	"					"
"	"	149	"	GND		OUT	IN	GND	"	"	"			"	"	"	2A to 2Y	"	"					"
"	"	150	"	"		OUT	GND	IN	"	"	"			"	"	"	2B to 2Y	"	"					"
"	"	151	"	"			"	GND	"	IN	"	OUT		"	"	"	4A to 4Y	"	"					"
"	"	152	"	"			"	"	"	GND	IN	OUT		"	"	"	4B to 4Y	"	"					"
"	"	153	"	"			"	"	"	"	GND		OUT	IN	"	"	3A to 3Y	"	"					"
"	"	154	"	"			"	"	"	"	"		OUT	GND	IN	"	3B to 3Y	"	"					"
t_{PLH}	"	155	IN	"	OUT		"	"	"	GND	GND			GND	GND	"	1A to 1Y	"	"					"
"	"	156	GND	IN	OUT		"	"	"	"	"			"	"	"	1B to 1Y	"	"					"
"	"	157	"	GND		OUT	IN	"	"	"	"			"	"	"	2A to 2Y	"	"					"
"	"	158	"	"		OUT	GND	IN	"	"	"			"	"	"	2B to 2Y	"	"					"
"	"	159	"	"			"	GND	"	IN	"	OUT		"	"	"	4A to 4Y	"	"					"
"	"	160	"	"			"	"	"	GND	IN	OUT		"	"	"	4B to 4Y	"	"					"
"	"	161	"	"			"	"	"	GND	GND		OUT	IN	"	"	3A to 3Y	"	"					"
"	"	162	"	"			"	"	"	GND	GND		OUT	GND	IN	"	3B to 3Y	"	"					"

See footnotes at end of device type 04.

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TABLE III. Group A inspection for device type 04 – Continued.

Symbol	MIL-STD-833 method	Case A,C,D,T,X,Y	Terminal conditions 1/														Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14		Subgroup 12 T_A = 25°C						
		Symbol	1A	1B	1Y	2Y	2A	2B	V_{SS}	4A	4B	4Y	3Y	3A	3B	V_{DD}								
		Test No.																Min	Max	Min	Max	Min	Max	
t_{THL}	3004	163	IN	GND	OUT		GND	GND	GND	GND	GND			GND	GND	10.0V	1Y	5	100					ns
"	Fig. 3	164	GND	IN	OUT		GND	GND	"	"	"			"	"	"	1Y	"	"					"
"	"	165	"	GND		OUT	IN	GND	"	"	"			"	"	"	2Y	"	"					"
"	"	166	"	"		OUT	GND	IN	"	"	"			"	"	"	2Y	"	"					"
"	"	167	"	"			"	GND	"	IN	"	OUT		"	"	"	4Y	"	"					"
"	"	168	"	"			"	"	"	GND	IN	OUT		"	"	"	4Y	"	"					"
"	"	169	"	"			"	"	"	"	GND		OUT	IN	"	"	3Y	"	"					"
"	"	170	"	"			"	"	"	"	"		OUT	GND	IN	"	3Y	"	"					"
t_{TLH}	"	171	IN	"	OUT		"	"	"	"	"			"	GND	"	1Y	"	"					"
"	"	172	GND	IN	OUT		"	"	"	"	"			"	"	"	1Y	"	"					"
"	"	173	"	GND		OUT	IN	"	"	"	"			"	"	"	2Y	"	"					"
"	"	174	"	"		OUT	GND	IN	"	"	"			"	"	"	2Y	"	"					"
"	"	175	"	"			"	GND	"	IN	"	OUT		"	"	"	4Y	"	"					"
"	"	176	"	"			"	"	"	GND	IN	OUT		"	"	"	4Y	"	"					"
"	"	177	"	"			"	"	"	GND	GND		OUT	IN	"	"	3Y	"	"					"
"	"	178	"	"			"	"	"	GND	GND		OUT	GND	IN	"	3Y	"	"					"

1/ Pins not designated may be "HIGH" level logic, "LOW" level logic or open. Exceptions are as follows: $V_{IC(pos)}$ tests, the V_{SS} terminals shall be open; $V_{IC(neg)}$ tests, the V_{DD} terminal shall be open; I_{SS} tests, the output terminals shall be open.
2/ The I_{SS} measurements shall be performed in sequence.
3/ The device manufacturer may, at his option, measure I_{IL} and I_{IH} at 25°C for each individual input or measure all inputs together.
4/ See 4.4.1c.

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TABLE IV. Delta limits at 25°C.

Parameter 1/	Device types			
	01	02	03	04
I_{SS}	±10 nA	±10 nA	±10 nA	±10 nA
I_{OL1}	±15%	±15%	±15%	±15%
I_{OH1}	±15%	±15%	±15%	±15%

1/ Each of the above parameters shall be recorded before and after the required burn-in and life tests to determine delta (Δ).

4.5.3 Quiescent supply current (I_{SS} test). When performing quiescent supply current measurements (I_{SS}), the meter shall be placed so that all currents flow through the meter.

4.5.4 Radiation hardness assurance (RHA) testing. The RHA testing shall be performed in accordance with test procedures and sampling specified in MIL-PRF-38535 and herein.

a. Before irradiation, selected samples shall be assembled in qualified packages and pass the governing electrical parameters (group A subgroup 1 at 25°C) and also be subjected to the threshold-voltage test in table VII in order to calculate the delta threshold (ΔV_T) after irradiation.

b. The devices shall be subjected to a total radiation dose as specified in MIL-PRF-38535 for the radiation hardness assurance level being tested, and meet the end-point electrical parameters as defined in table V at 25°C, after exposure. The start and completion of the end-point electrical parameter measurements shall not exceed 2 hours following irradiation.

c. Threshold-voltage test circuit conditions shall be as specified in table VII and figure 4. In situ and remote testing, the tests shall be performed with the devices biased in accordance with table VI and the bias may be interrupted for up to 1 minute to remove devices to the remote bias fixture.

d. After irradiation, the devices shall pass the truth table test as specified in subgroup 7 in table III or if subgroup 7 is not required, then an equivalent truth table test shall be performed.

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TABLE V. Radiation hardened end-point electrical parameters at 25°C.

Parameter	All device types	V_{DD}
V_{TN}	0.3 V min	10 V
V_{TP}	2.8 V max	10 V
ΔV_T	1.4 V max	10 V
I_{SS}	100 x max limit	18 V
t_{PLH}	1.35 x max limit	5 V
t_{PHL}	1.35 x max limit	5 V

TABLE VI. Bias during exposure to radiation.

Device type	Pin connections 1/		
	V_{DD} = 10 V dc (through a 30 kΩ to 60 kΩ resistor)	V_{SS} = GND	V_{DD} = 10 V dc
01	1, 2, 5, 6, 8, 9, 10, 11, 12, 13	7	14
02	1, 2, 5, 6, 8, 9, 10, 11, 12, 13	7	14
03	1, 2, 5, 6, 8, 9, 12, 13	7	14
04	1, 2, 5, 6, 8, 9, 12, 13	7	14

1/ Pins not designated are open, or tied to 10 V dc through a 30 kΩ to 60 kΩ resistor.

4.6 Data reporting. When specified in the purchase order or contract, a copy of the following data, as applicable, shall be applied.

a. Attributes data for all screening tests (see 4.2) and variables data for all static burn-in, dynamic burn-in, and steady-state life tests (see 3.6).

b. A copy of each radiograph.

c. The technology conformance inspection (TCI) data (see 4.4).

d. Parameter distribution data on parameters evaluated during burn-in (see 3.6).

e. Final electrical parameters data (see 4.2d).

f. RHA delta limits.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency, or within the military service's system command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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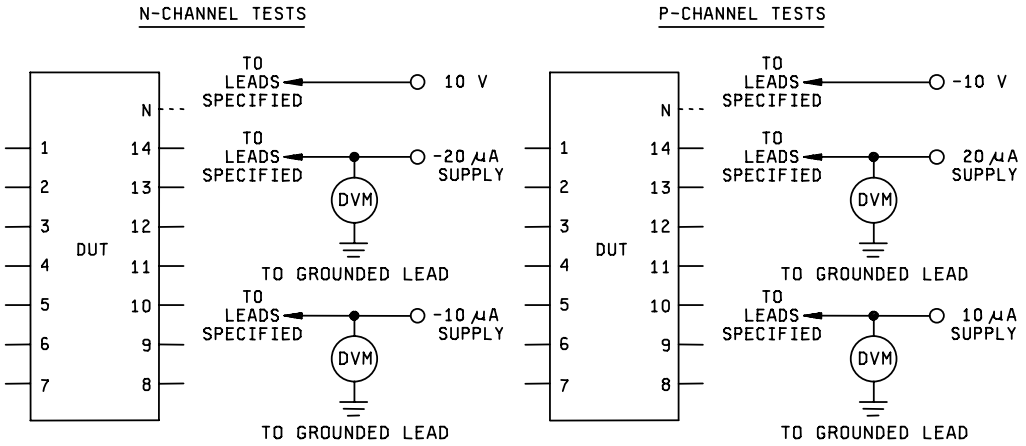


FIGURE 4. Threshold-voltage test circuit.

TABLE VII. Threshold-voltage test circuit conditions.

Device	GND	10 V	V_{TN} measured at		GND	-10 V	V_{TP} measured at	
			-20 µA supply	-10 µA supply			20 µA supply	10 µA supply
01	1	2, 14		5-13	1	5-13		2, 14
02	1	2, 14		5-13	1	5-13		2, 14
03	1	14		2, 5-9, 12, 13	1	2, 5-9, 12, 13		14
04	1	14		2, 5-9, 12, 13	1	2, 5-9, 12, 13		14

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6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. PIN and compliance identifier, if applicable (see 1.2).

c. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

d. Requirements for certificate of compliance, if applicable.

e. Requirements for notification of change of product or process to contracting activity in addition to notification to the qualifying activity, if applicable.

f. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

g. Requirements for product assurance and radiation hardness assurance options.

h. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements should not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

i. Requirements for "JAN" marking.

j. Packaging requirements. (see 5.1)

6.3 Superseding information. The requirements of MIL-M-38510 have been superseded to take advantage of the available Qualified Manufacturer Listing (QML) system provided by MIL-PRF-38535. Previous references to MIL-M-38510 in this document have been replaced by appropriate references to MIL-PRF-38535. All technical requirements now consist of this specification and MIL-PRF-38535. The MIL-M-38510 specification sheet number and PIN have been retained to avoid adversely impacting existing government logistics systems and contractors parts lists.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List QML-38535 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DSCC-VQ, P.O Box 3990, Columbus, Ohio 43218-3990.

6.5 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-PRF-38535, MIL-HDBK-1331, and as follows:

C_i ..... Input terminal-to-GND capacitance.
GND ..... Ground zero voltage potential.
T_A ..... Free air temperature.
V_{IC} ..... Input clamp voltage.
V_{DD} ..... Positive supply voltage.
V_{SS} ..... Negative supply voltage.
I_{SS} ..... Quiescent supply current.

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6.6 Logistic support. Lead materials and finishes (see 3.4) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class S for National Aeronautics and Space Administration or class B for Department of Defense (see 1.2.2), lead material and finish A (see 3.4). Longer length leads and lead forming should not affect the part number.

6.7 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges, post irradiation performance or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information should not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-PRF-38535.

Military device type	Generic-industry type
01	4085B
02	4086B
03	4070B
04	4077B

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5962-2029)

Review activities:
Army - MI, SM
Navy - AS, CG, MC, SH, TD
Air Force – 03, 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST Online database at www.dodssp.daps.mil.

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